

Lierda NT26KxBx series Hardware Design Manual

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Revision History of the Document

Document version	Change date	Reviser	Auditor	Change content
Rev1.0	25-09-28	CWY、ZH	YMX、SLY	Initial version
Rev1.1	25-10-29	CWY、ZH	YMX、SLY	Add descriptions for NT26K2B1-0S-VN modules.
Rev2.0	25-11-11	CWY、ZH	YMX、SLY	Adjust document name

Safety Instructions

Users are responsible for complying with the relevant regulations of other countries regarding wireless communication modules and devices, as well as specific environmental regulations. By following the following safety principles, personal safety can be ensured and help protect products and work environments from potential damage. Our company is not responsible for any related losses caused by customers' failure to comply with these regulations.



Safety comes first when driving! Do not use handheld mobile devices while driving, unless they have a hands-free function. Please pull over before making a call!



Please turn off your mobile terminal device before boarding. The wireless function of the mobile terminal is prohibited from being turned on in the aircraft to prevent interference with the aircraft communication system. Ignoring this prompt may compromise flight safety and even violate the law.



When in a hospital or health care facility, pay attention to any restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so it may be necessary to turn off mobile terminal devices.



Mobile terminal devices do not guarantee effective connection in all situations, such as when there is no credit on the mobile terminal device or the SIM card is invalid. In case of emergency when encountering the above situations, remember to use emergency calls, ensuring that your device is powered on and in an area with sufficient signal strength.



Your mobile terminal device will receive and transmit radio frequency signals when it is powered on, which may cause radio frequency interference when it is close to a TV, radio, computer, or other electronic devices.



Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off the mobile terminal device. Operating electronic devices in any potentially explosive hazardous location poses a safety risk.

Module selection for application.

Serial Number	Module model	Support frequency band	Explanation
1	NT26K2B1-0S-VN	Band1/3/5/8/ 34/38/39/40/41*	VDD_EXT sleep mode power off, main chip changed to 716E, lowest power consumption supports HIB mode, single card version

Notes

LTE-TDD B41 only supports 140M (2535-2675MHz).

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1 Introduction

This document defines the application specifications of the Lierda NT26KxBx series Cat.1 wireless communication module, describing its frequency band function, key features, hardware interface, operating modes, electrical characteristics, mechanical specifications, packaging storage, etc., to help users quickly grasp the application method of this Cat.1 module and quickly and flexibly carry out product design.

1.1 Special characters

Table 1-1 Special Symbol Description

Symbol	Definition
*	If not specifically stated, the asterisk (*) marked after the module function, feature, interface, pin name, AT command, or parameter indicates that the function, feature, interface, pin name, AT command, or parameter of the module is under development and not currently supported.

2 Product Overview

The NT26KxBx series module is a leading Cat.1 wireless communication module globally, compliant with 3GPP R14 Cat.1bis standard, supporting bandwidths of 1.4/3/5/10/15/20MHz, featuring small size, low power consumption, strong anti-interference capability, etc. The module is suitable for various common IoT application scenarios, such as:

- ◆ Smart meter reading
- ◆ Intelligent Parking
- ◆ Smart City
- ◆ Intelligent Security
- Asset tracking
- ◆ Smart home appliances
- ◆ Agriculture and environmental monitoring, etc.

2.1 Key Features

The key features of the NT26KxBx series module are as follows:

Table 2-2 Module Key Features Description

Parameters	Explanation
Module packaging	LCC+LGA
Module dimensions	17.7mm×15.8mm×2.4mm (L×W×H)
Module weight	1.3g
Operating voltage	◆ Power supply voltage: 3.3 to 4.5V, typical value: 3.8V ◆ Extended power supply voltage (1): 3.1~4.5V
VDD_EXT characteristics	The NT26KxBx series offers two hardware versions. Sleep mode: VDD_EXT power off Sleep mode: VDD_EXT power on
Ultra-low power	Shutdown mode: ≤2uA

Transmit power	23dBm±2.7dB (Max)
Antenna interface	50Ω characteristic impedance
USIM	Supported USIM card types: Class B (3.0 V) and Class C (1.8 V), see section 4.3 for details.
UART	Main Serial Port 1: ◆ Used for AT command transmission and data transfer, the supported baud rate defaults to 115200bps. ◆ Used for firmware upgrade, the supported baud rate defaults to 921600bps.
	Debugging serial port 1: ◆ Used for software debugging. ◆ The default baud rate of the serial port is 3Mbps.
	1st auxiliary serial port: ◆ User-defined function
USB	◆ 1 USB port ◆ Compatible with USB 2.0 (supporting slave mode only), with a maximum data transfer rate of 480 Mbps. ◆ Used for AT command transmission, data transfer, software debugging, and firmware upgrade.
I2C	◆ 1 I2C Interface ◆ Compliant with I2C bus protocol specifications ◆ Support standard mode 100KHz, fast mode 400KHz
SPK	◆ 1 channel analog audio interface: ◆ Used for audio, can directly drive low-power headphones or connect to an external PA to drive speakers.
Indicator light	◆ Running status indicator ◆ Network status indicator
Communication Interface Characteristics	Support 3GPP Rel.13/14 Cat.1 wireless communication interface and protocol.
Network Protocol Features (2)	Support protocols such as TCP/UDP/HTTP/SSL*/MQTT/FOTA/PPP*/RNDIS*/FTP*/ECM* and so on.
Data transmission characteristics	◆ Maximum support for Cat.1 FDD and TDD ◆ 1.4/3/5/10/15/20MHz RF bandwidth

	◆ Upstream QPSK, 16QAM ◆ Downlink QPSK, 16QAM, 64QAM ◆ LTE-FDD: Maximum downlink speed of 10Mbps, maximum uplink speed of 5Mbps ◆ LTE-TDD: maximum downlink speed of 8.96Mbps, maximum uplink speed of 3.1Mbps
Firmware upgrade	◆ Main serial port upgrade ◆ USB interface upgrade ◆ DFOTA upgrade
Operating temperature	◆ Normal operating temperature range (3): -35~75℃ ◆ Extend operating temperature range (4): -40~85℃
Storage temperature	Storage Temperature Range(5): -40~90℃
RoHS	All components comply with the EU RoHS standards.

Note

(1) Refer to section 6.2 for a detailed description of the extended voltage.

(2) Different firmware versions may support different network protocols, please refer to the specific firmware version documentation for details.

(3)(4)(5) See section 6.6 for specific descriptions about temperature.

2.2 Function block diagram

The NT26KxBx series module has rich peripheral interfaces and RF functions. The functional block diagram is as follows:

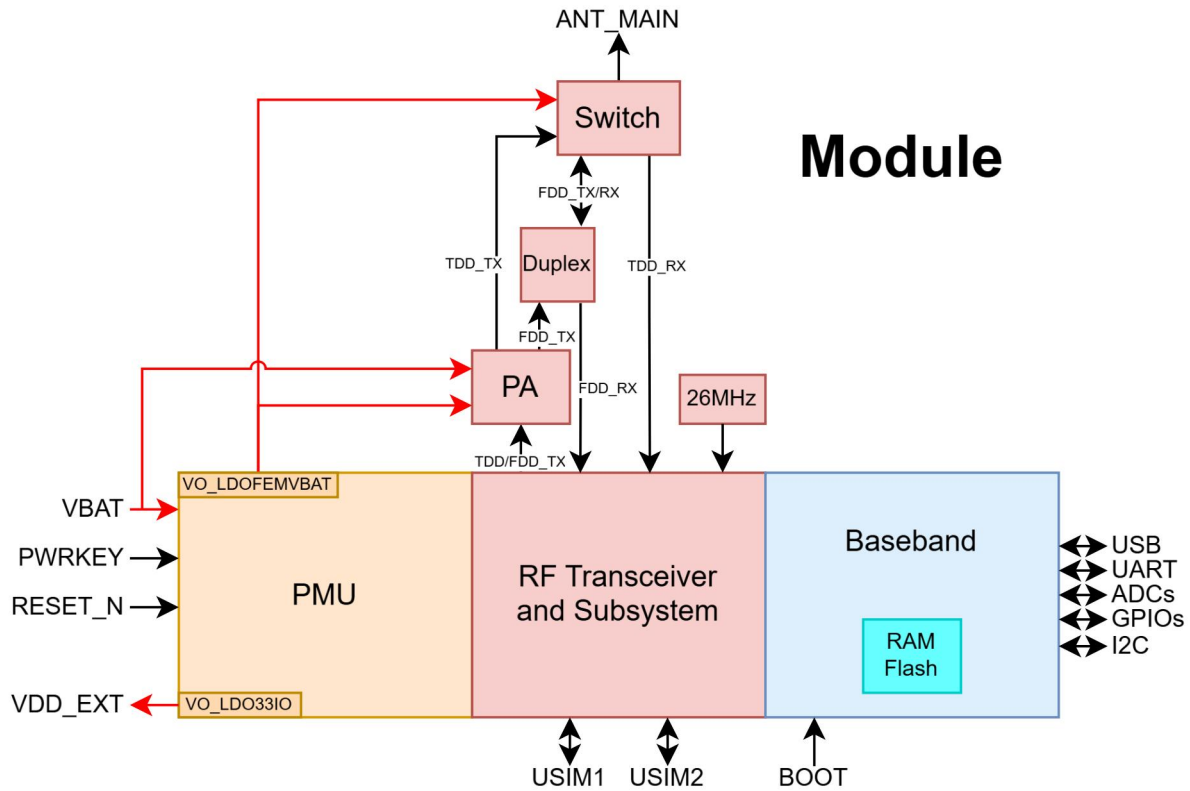


Figure 2.1 Module Functional Block Diagram

2.3 Pinout diagram

The NT26KxBx series module has a total of 109 pins, including 48 LCC pins and the remaining 61 LGA pins. The pin distribution diagram of the module is as follows:

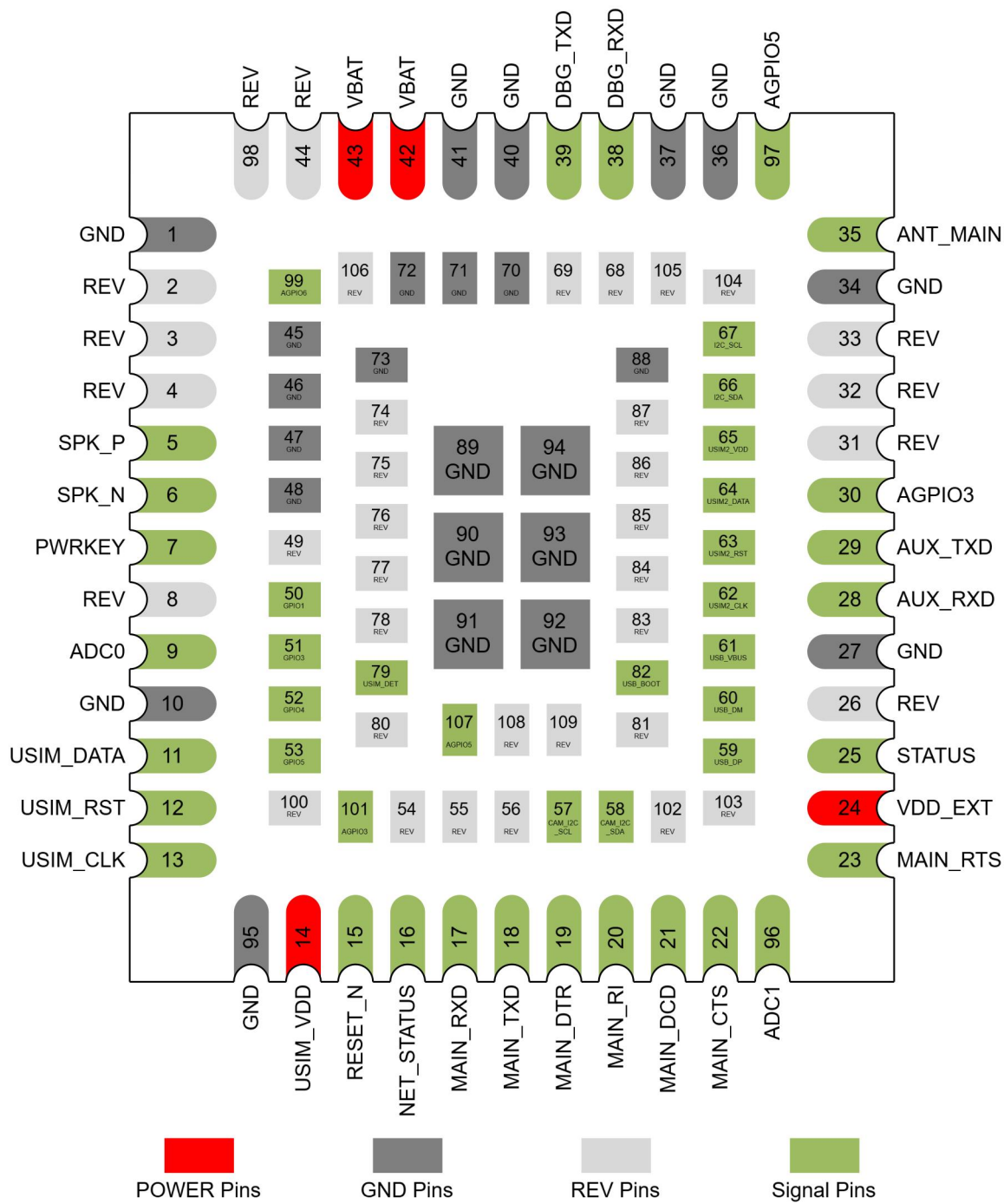


Figure 2.2 Module Pinout Diagram

2.4 Pin Description Table

In order to better understand the application, the following table explains the type definition of I/O parameters:

Table 2-4 Description of I/O Parameter Type Definitions

I/O parameter types	Explanation
DI	Numeric input
DO	Numeric output
DIO	Numeric input and output
AI	Simulated input
AO	Simulated output
AIO	Simulated input and output
PI	Power input
PO	Power output
G	GND
OD	Leaking pole opens the way

To better understand the application, the following table provides a description of the characteristics of the power domain parameters:

Table 2-5 Module Power Domain Characteristics Description

Power domain parameters Type	DC characteristics	Explanation	Power supply object
VBAT	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	Module power input, recommended to use 3.8V/1.2A power supply.	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 1.8V; Different hardware versions: In sleep mode, VDD_EXT power-off version, driving capability is 120mA; In sleep mode, the non-power-off version of VDD_EXT has a driving capability of 3mA.	VDD_EXT
VO_LDOIO	VOLmax=0.2×VO_LDOIO VOHmin=0.7×VO_LDOIO VILmax=0.15×VO_LDOIO VIHmin=0.8×VO_LDOIO	Default 1.8V, configurable by software; Power off in sleep mode.	UART GPIO

VDD18AON	VILmax=0.2×VDD18AON VIHmin =0.7×VDD18AON VOLmax=0.15×VDD18AON VOHmin=0.8×VDD18AON	Does not lose power in sleep mode.	WAKEUP RESET
LDO_AONIO	VILmax=0.2×LDO_AONIO VIHmin =0.7×LDO_AONIO VOLmax=0.15×LDO_AONIO VOHmin=0.8×LDO_AONIO	Default 1.8V, configurable by software; No power consumption in sleep mode.	AGPIO
VO_LDOSIM	Vnorm=1.8/3.0V VOLmax=0.15×VO_LDOSIM VOHmin=0.8×VO_LDOSIM VILmax=0.2×VO_LDOSIM VIHmin=0.7×VO_LDOSIM	SIM card dedicated power supply, supporting cards with 1.8/3.0V.	USIM

Note

◆ Power Domain: A region in a chip or system that is divided into independently powered areas, where each area can individually control the power on/off, voltage level, or power consumption mode.

◆ VDD_EXT and VO_LDOIO are the same power supply.

Table 2-6 Module Pin Description

1. Power					
Pin number	Pin names	I/O	Describe	DC characteristics	Note
42,43	VBAT	PI	Power supply	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	External power supply needs to provide a current carrying capacity of 1.2A or above.
24	VDD_EXT	PO	Power output	Vnorm=1.8V Iomax=120mA/3mA	Can be used for external circuit pull-up or reference voltage; there are two modes of power loss and non-power loss in sleep mode; if not

					used, leave floating, it is recommended to reserve test points.
1,10,27,34, 36~37,40~41,45~4 8,70~72,73,88~95	GND	G	GND	-	-

2. Power on/off and reset

Pin number	Pin names	I/O	Description	DC characteristics	Note
7	PWRKEY	DI	Module power on/off pin	$V_{ILmax}=0.5V$	Internal pull-up, low level effective.
15	RESET_N	DI	Module reset pin	$V_{ILmax}=0.5V$	Internal pull-up, low level effective.

3. USB

Pin number	Pin names	I/O	Description	DC characteristics	Note
59	USB_DP*	DIO	USB differential data (+)	-	Compliant with USB 2.0 specifications. Requires 90 Ω differential impedance. Test points need to be reserved. If not used, leave floating. (Default not enabled)
60	USB_DM*	DIO	USB differential data (-)	-	
61	USB_VBUS*	DI	USB_VBUS input wake-up	$V_{max} = 5.25 V$ $V_{min} = 3.0 V$ $V_{nom} = 5.0 V$	

4. Serial Port UART

Pin number	Pin names	I/O	Description	DC characteristics	Note
------------	-----------	-----	-------------	--------------------	------

r				tics	
17	MAIN_RXD	DI	Main serial port receives data	VO_LDOIO	If not used, leave floating
18	MAIN_TXD	DO	The main serial port sends data.	VO_LDOIO	If not used, leave floating
19	MAIN_DTR	DI	Main serial port data terminal ready	VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.
20	MAIN_RI	DO	Main serial port outputs ringing prompt	LDO_AONIO /VDD18AON	Default high level, output low level for 120ms when the module needs to report URC, and restore high level after data output is completed.
21	MAIN_DCD*	DO	Main serial port output carrier detection	VO_LDOIO	Internally connected in parallel with pin 66, cannot be used simultaneously.
22	MAIN_CTS*	DO	Main serial port DTE clear to send	VO_LDOIO	Internally connected in parallel with pin 57, cannot be used simultaneously.
23	MAIN_RTS*	DI	The main serial port DTE requests to send.	VO_LDOIO	Internally connected in parallel with pin 58, cannot be used simultaneously
38	DBG_RXD	DI	Debugging serial	VO_LDOIO	If not used, leave

			port to receive data		floating
39	DBG_TXD	DO	Debug serial port to send data	VO_LDOIO	If not used, leave floating
28	AUX_RXD	DI	Assist in receiving data through the serial port.	VO_LDOIO	Connected in parallel with pin 52 internally, cannot be used simultaneously.
29	AUX_TXD	DO	Assist in sending data through the serial port.	VO_LDOIO	Connected in parallel with pin 53 internally, cannot be used simultaneously.

5. External USIM card interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
11	USIM_DATA	DIO	SIM card data cable	VO_LDOSIM	
12	USIM_RST	DIO	SIM card reset line	VO_LDOSIM	
13	USIM_CLK	DIO	SIM card clock line	VO_LDOSIM	
14	USIM_VDD	PO	SIM card power	VO_LDOSIM	1.8/3.0V
62	USIM2_CLK	DIO	SIM2 (secondary card) clock line	VO_LDOIO	
63	USIM2_RST	DIO	SIM2 (secondary card) reset line	VO_LDOIO	Internally connected in parallel with pin 50, cannot be used simultaneously.
64	USIM2_DATA	DIO	SIM2 (secondary card) data cable	VO_LDOIO	Internally connected in parallel with pin 51, cannot

					be used simultaneously.
65	USIM2_VDD	PO	SIM2 (secondary card) power	VO_LDOSIM	Powered by the same source as USIM_VDD
79	USIM_DET	DI	SIM card status signal detection	VDD18AON	High level effective

6. Antenna interface RF

Pin number	Pin names	I/O	Description	DC characteristics	Note
35	ANT_MAIN	AIO	RF antenna interface	-	50Ω characteristic impedance

7. I2C interface

Pin number	Pin names	I/O	Describe	DC characteristics	Note
57	CAM_I2C_SCL	DO	I2C serial clock	VO_LDOIO	Internally connected in parallel with pin 22, cannot be used simultaneously.
58	CAM_I2C_SDA	DIO	I2C serial data	VO_LDOIO	Internally connected in parallel with pin 23, cannot be used simultaneously.

66	I2C_SDA	DIO	I2C serial data	VO_LDOIO	Connected internally in parallel with pin 21, cannot be used simultaneously; used as I2C data signal, an external 4.7K pull-up resistor needs to be added, leave floating if not used.
67	I2C_SCL	DO	I2C serial clock	VO_LDOIO	I2C clock signal, need to add an external 4.7K pull-up resistor, leave floating if not used.

8. Simulation audio interface *					
Pin number	Pin Names	I/O	Describe	DC Characteristics	Note
5	SPK_P	DO	Audio differential output	VO_LDOIO	Internally connected in parallel with pin 82, cannot be used simultaneously.
6	SPK_N	DO	Audio negative differential output	Internal Ground	If not used, leave floating

9、ADC interface					
Pin number	Pin names	I/O	Description	DC characteristics	Note
9	ADC0	AI	Universal ADC interface	Voltage range: 0~1.05V	If not used, leave floating
96	ADC1	AI	General ADC interface		If not used, leave floating

10. Status Indicator					
Pin number	Pin names	I/O	Description	DC characteristics	Note
16	NET_STATUS	DO	Network status indicator	VDD18AON	If not used, leave floating
25	STATUS	DO	Operating status indicator	LDO_AONIO /VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep

11. Other interfaces					
Pin number	Pin names	I/O	Description	DC characteristics	Note
82	USB_BOOT	DI	Emergency download mode control. Pull up to VDD_EXT before the module is powered on, and the module will enter emergency download mode	VO_LDOIO	Internally connected in parallel with pin 5, cannot be used simultaneously;

			after startup.		The module defaults to weak pull-down internally, with high level active. Before the module starts up normally, it is forbidden to pull up this pin. It is recommended to reserve a test point.
50	GPIO1	DIO	General-purpose GPIO	VO_LDOIO	Connected in parallel with pin 63 internally, cannot be used simultaneously.
51	GPIO3	DIO	General Purpose Input/Output (GPIO)	VO_LDOIO	Internally connected in parallel with pin 64, cannot be used simultaneously.
52	GPIO4	DIO	General Purpose Input/Output (GPIO)	VO_LDOIO	Connected in parallel with pin 28 internally, cannot be used simultaneously.
53	GPIO5	DIO	General Purpose Input/Output (GPIO)	VO_LDOIO	Internally connected in parallel with pin 29, cannot be

					used simultaneously
97,107	AGPIO5	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating
99	AGPIO6	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	Currently serving as the 62nd foot internal pull-up power supply, external use is prohibited.
30,101	AGPIO3	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating

12. Reserved interface

Pin number	Pin names	I/O	Description	DC characteristics	Note
2~4,8,26,31~33,44,49,54~56,74,74~78,80~81,83~87,98,100,102~105,106,108~109	REV	-	-	-	Undefined pins, suggest floating, not recommended for customer use (pins 68, 69, 75, 86, 87, 105 are prohibited from use, handle floating).

Note: "*" indicates the feature is under development.

Note

1. AGPIO driving capability is limited, high-level output is prone to fluctuation, only suitable for signal level identification, cannot be used as an external power supply.

2. It is recommended to leave PIN8, PIN68, PIN69, PIN75, PIN86, PIN87, PIN105 pins

floating, and grounding is prohibited.

3. AGPIOWU driving capability is less than 30uA, can only be used as an input, not as an output.

PIN50 and PIN51 pins are not recommended for use, it is recommended to leave them floating.

2.5 Evaluation suite

Lierda can provide a complete evaluation and development kit, including module adapter boards, Cat.1 & NB-IoT module universal development boards, etc. Please feel free to contact us for consultation.

3 Work Characteristics

3.1 Working mode

The NT26KxBx series module has multiple operating modes. The following table briefly describes several common operating modes of the module:

Table 3-1 Working Modes

Pattern	Status Description
Normal working mode	Data transmission: Network connection is normal. Module power consumption depends on network environment and data transfer rate.
	Idle status: The software is running normally. The module is connected to the network but not exchanging data with the network.
Minimum Function Mode	AT+CFUN=0 can set the module to minimum functionality mode, where both the RF and USIM card are not working.
Sleep mode	The module's power consumption will be reduced to a minimum, but the module can still receive paging, short messages, and TCP/UDP data.
Shutdown mode	VBAT power supply is not disconnected, software stops working

Note

1. Please refer to the "AT Command Manual" for related AT+CFUN commands.
2. Detailed working mode and corresponding power consumption data reference PMU application document.

3.2 Sleep mode

The module can enter sleep mode via AT commands. In sleep mode, the module can reduce power consumption to a very low level while still being able to receive network information through RF.

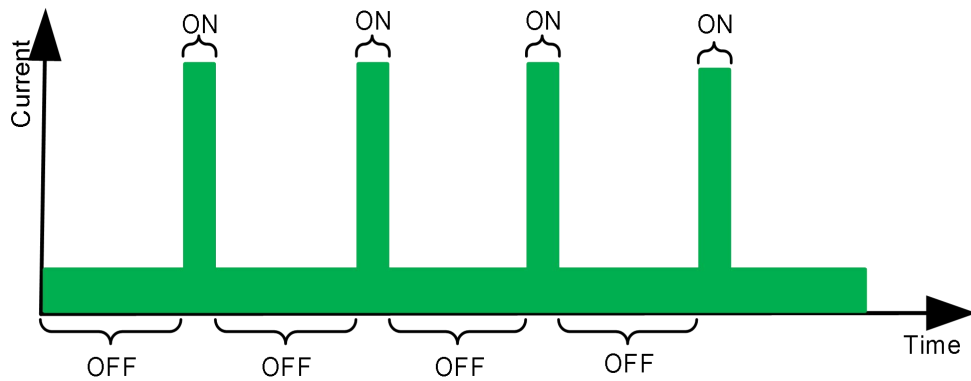


Figure 3.1 Power Consumption States of the Module in Sleep Mode

In sleep mode, the module's peripherals are powered off to reduce power consumption, but the RF is in intermittent reception state and can still receive network information. When the module has a URC to report, it will wake up the host through the MAIN_RI pin.

3.2.1 Main serial port application scenarios

If the module and the host communicate via UART, two conditions must be met simultaneously to put the module into sleep mode:

- ◆ Executing the command `AT+QSCCLK=1` or `AT+LSCLK=1` (please refer to the "AT Command Manual" for specific commands) enables the sleep function;

Ensure that MAIN_DTR remains at a high level or floating.

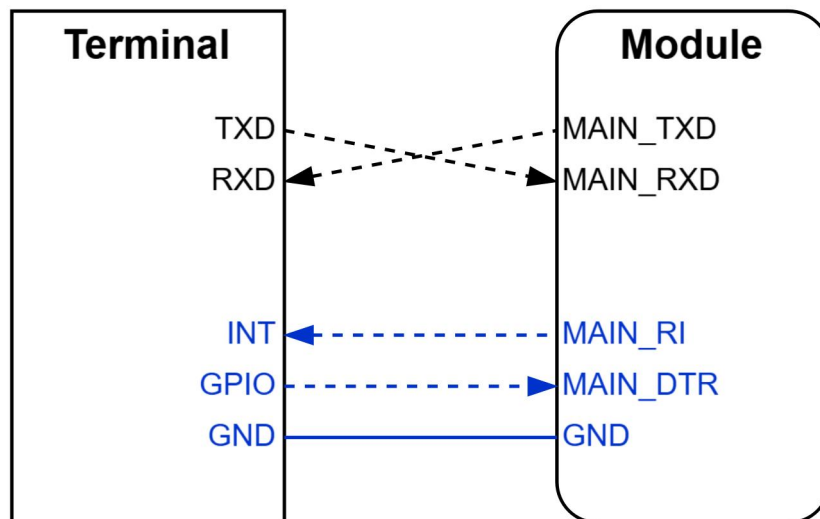


Figure 3.2 Application Circuit of Sleep Mode UART

When the module needs to report URC, the module notifies the host through MAIN_RI.

- ◆ The host pulling down MAIN_DTR can wake up the module.

If the host continues to pull down MAIN_DTR, it can prevent the module from entering sleep mode.

If the sleep function is enabled using AT+QSCLK=2 or AT+LSCLK=2 (please refer to the "AT Command Manual" for specific commands), the module can be woken up by sending any command to the main UART interface.

3.2.2 USB application scenarios*

In the USB application scenario, the module needs to meet the following three conditions to enter sleep mode:

- ◆ Perform AT+QSCLK=1/AT+LSCLK=1 (select according to the "AT Command Manual");
- ◆ Ensure that MAIN_DTR remains at a high level or floating;
- ◆ The host USB bus connected to the module's USB interface enters the suspend state.

3.2.2.1 Support USB remote wake-up function.

The host supports USB suspend and resume, as well as USB remote wake-up functionality.

Please refer to the diagram for the connection between the module and the host:

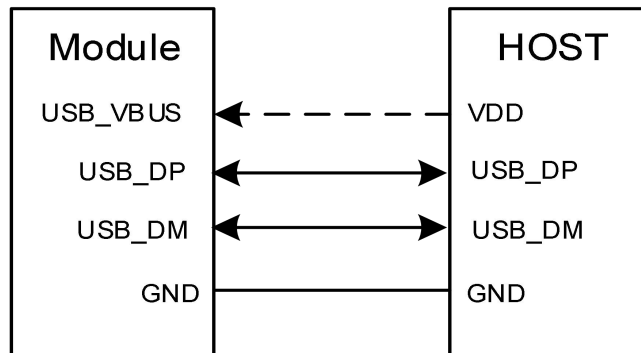


Figure 3.3 Sleep Application with USB Remote Wake-up Function

- ◆ Sending data to the module via USB will wake up the module.

When the module has URC reporting, the module will send a remote wake-up signal via the USB bus to wake up the host.

3.2.2.2 Support USB suspend and resume as well as MAIN_RI function*

If the host supports USB suspend and resume but does not support USB remote wake-up function, the host needs to be woken up by the MAIN_RI signal of the module.

The connection between the module and the host reference the diagram below:

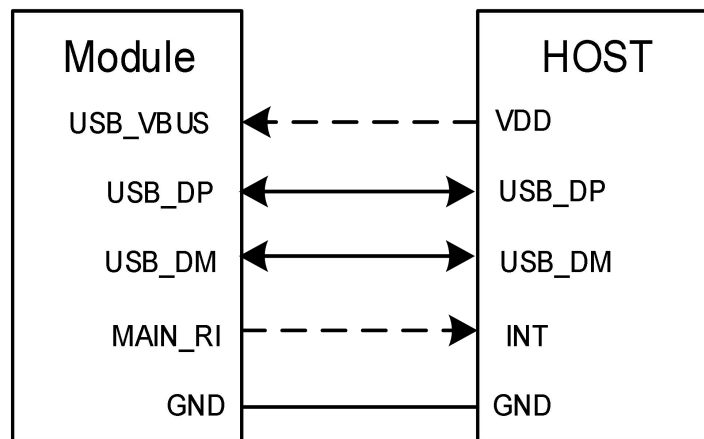


Figure 3.4 Sleep Application with MAIN_RI Function

- ◆ Sending data to the module via USB will wake up the module;
- ◆ When the module has URC report, the module can wake up the host through MAIN_RI.

3.2.2.3 Does not support USB suspend function.

When the host does not support USB suspend function, the module needs to meet the following three conditions to enter sleep mode:

- ◆ Execute AT+QSCLK=1 or AT+LSCLK=1 (please refer to the "AT Command Manual")

for specific commands);

- ◆ Ensure that MAIN_DTR remains at a high level or floating;
- ◆ Disconnect the USB_VBUS power supply.

If the host does not support USB suspend function, the module can enter sleep mode by disconnecting USB_VBUS through an external control circuit.

The connection between the module and the host refers to the following diagram:

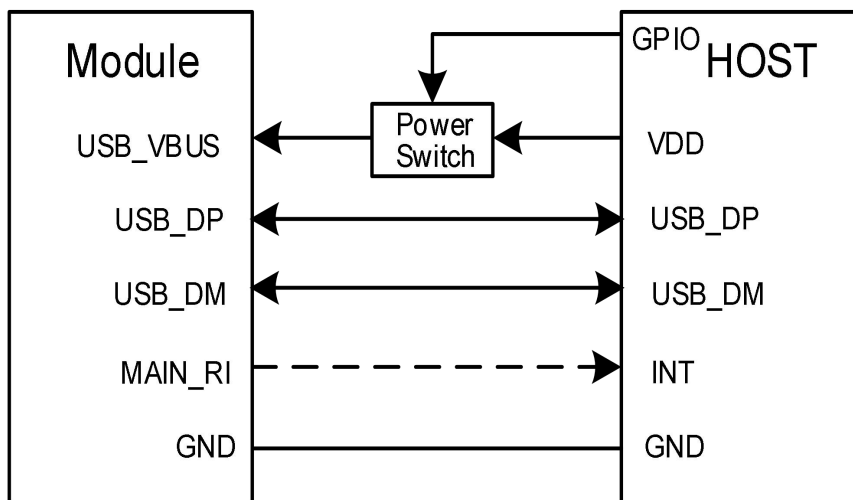


Figure 3.5 Sleep application that does not support USB suspend functionality

Note

Restoring USB_VBUS power supply can wake up the module.

3.3 Flight mode

When the module enters flight mode, the RF function is not available, and all AT commands related to RF cannot be accessed. The module can enter flight mode in the following ways:

The settings can be configured by sending the AT+CFUN=<fun> command. The <fun> parameter can be set to 0 or 1.

- ◆ AT+CFUN=0: Minimum functionality mode (turn off RF function and USIM card).
- ◆ AT+CFUN=1: Full functionality mode (default).

3.4 Power supply design

3.4.1 Power Introduction

The NT26KxBx series module provides 2 VBAT pins for connecting external power sources, and VDD_EXT is the module's external power output that can be used as a reference voltage. The interface description is as follows:

Table 3-2 Power Supply Pin Definitions

Pin number	Pin names	Description	Minimum value	Typical value	Maximum value	Unit
42,43	VBAT	Module power	3.3	3.8	4.5	V
24	VDD_EXT	Power output	-	1.8	-	V

The VBAT power supply range of the module is 3.3 to 4.5V, and it is necessary to ensure that the minimum input voltage is not lower than 3.3V when the module is operating. The following figure illustrates the VBAT voltage drop situation of the module during sudden transmission:

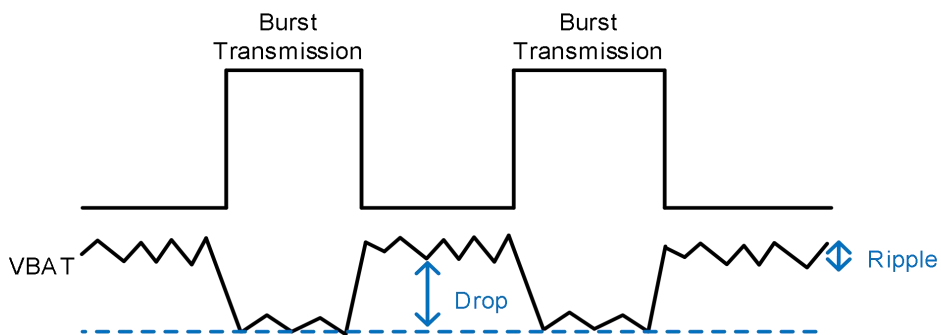


Figure 3.6 Emergency Transmission Power Requirements

Note

In sleep mode, VDD_EXT will power off.

3.4.2 Power supply design

When the module is operating at the maximum transmission power in a 4G network, the transient operating current under the current network may reach 1.2A, which may cause a drop in the power supply voltage. In any case, it is necessary to ensure that the module's power supply voltage remains above 3.3V, otherwise the module may experience unexpected conditions such as rebooting. To reduce voltage drops, it is recommended to use a low ESR (ESR $<0.7\Omega$) 100uF filtering capacitor. It is also recommended to reserve three chip multilayer ceramic capacitors (MLCC) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Additionally, it is recommended to add a high-power, low-clamping voltage, and high reverse pulse current I_{pp} TVS diode at the VBAT terminal to improve the module's surge voltage withstand capability.

The power supply reference circuit is as follows:

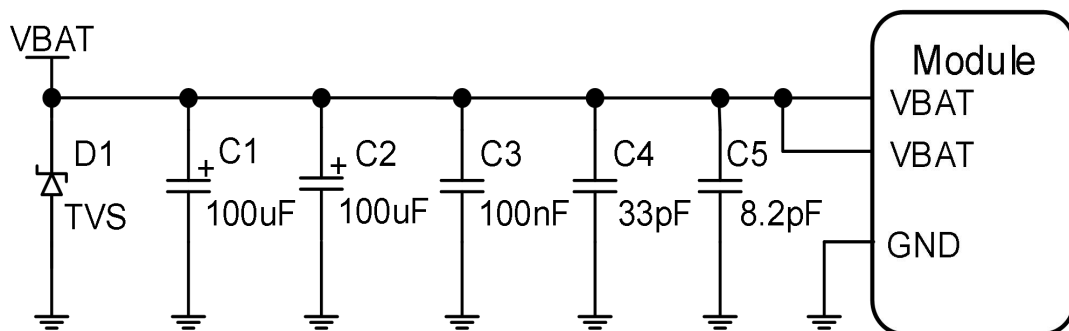


Figure 3.7 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider the trace width should be. It is recommended that the trace width should not be less than 2mm. The GND plane of the power supply section should be as complete as possible and have multiple GND vias. Also, decoupling capacitors should be placed as close as possible to the VBAT pin of the module.

- ◆ D1 is a TVS diode with high power and low clamping voltage;
- ◆ C1, C2 are 100uF low ESR tantalum capacitors, which enhance the power supply's ripple current capability and stabilize the voltage.
- ◆ C3~C5 are ceramic packaged filter capacitors of 100nF, 33pF, and 8.2pF respectively, used to eliminate high-frequency interference.

3.5 Power on.

3.5.1 Power-on introduction

The module can be powered on through the PWRKEY pin, the pin description is as follows:

Table 3-3 PWRKEY Pin Description

Pin number	Pin names	I/O	Description	Note
7	PWRKEY	DI	Module power on, low level active	Default to high level

The control logic of PWRKEY is as follows:

- ◆ The module can be powered on by pulling down the PWRKEY for at least 1 second when it is turned off;

3.5.2 Power-on circuit reference design

It is recommended to use an open-drain drive circuit or directly control the power-on through a button. To prevent electrostatic discharge generated by contact, a TVS should be placed near the button for ESD protection. Please refer to the circuit below:

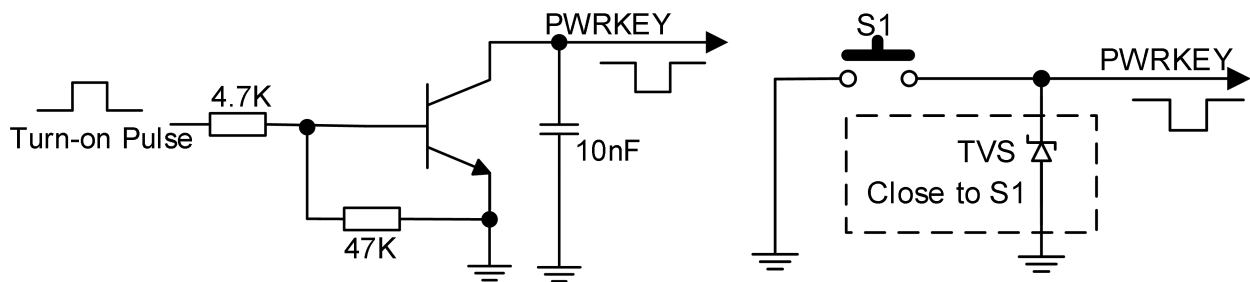


Figure 3.8 Power On/Off Circuit

3.5.3 Power-on sequence

The power-on sequence of the module is as shown in the figure below:

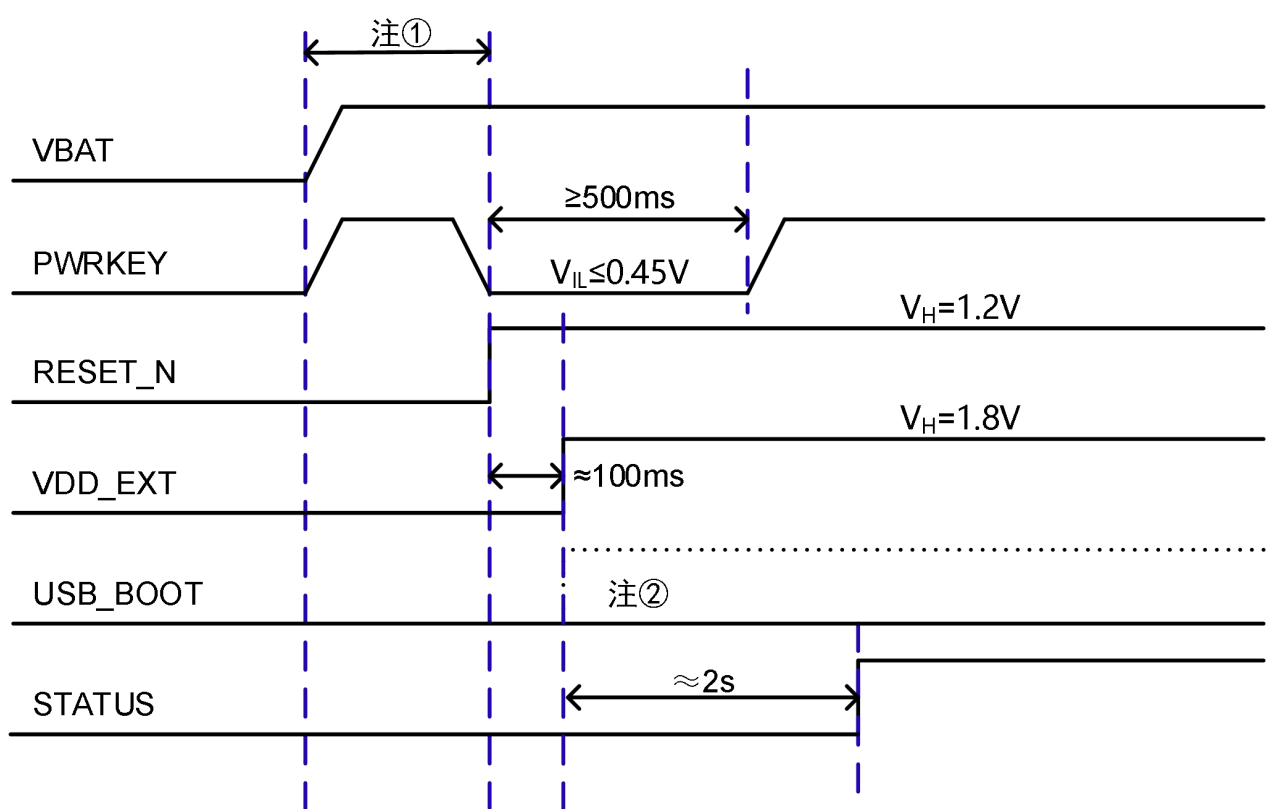


Figure 3.9 Power-on Boot Sequence Diagram

Notes

◆ The initial state of VBAT when powered on needs to be less than 0.1V, and the time for VABT to rise from 0V to 2.5V should be less than 10ms; before pulling down the PWRKEY pin, ensure that the VBAT voltage is stable, and it is recommended that the time interval between powering on VBAT and pulling down the PWRKEY pin is not less than 30ms.

◆ Pulling the USB_BOOT pin to VDD_EXT before powering on the module will enter emergency download mode when the module is powered on.

◆ If automatic power-on is required without the need for shutdown function, PWRKEY can be directly pulled down to GND, and the pull-down resistor is recommended to be 4.7k Ω , or PWRKEY can be controlled to low level by GPIO before powering on the module and keep it low.

◆ The main serial port of the module sends the characters "Lierda" or "RDY", which means the boot-up process is completed, please refer to the "AT Command Manual" for details.

3.6 Power off.

The module can be shut down in the following ways:

◆ Power off the module by controlling it through the PWRKEY pin.

◆ Send the command AT+QPOWD or AT+LPOWD (please refer to the "AT Command Manual" for specific commands) to power off.

3.6.1 PWRKEY shutdown

When the device is powered on, pulling down PWRKEY for 650ms and then releasing it will trigger the module to shut down. The timing sequence is as follows:

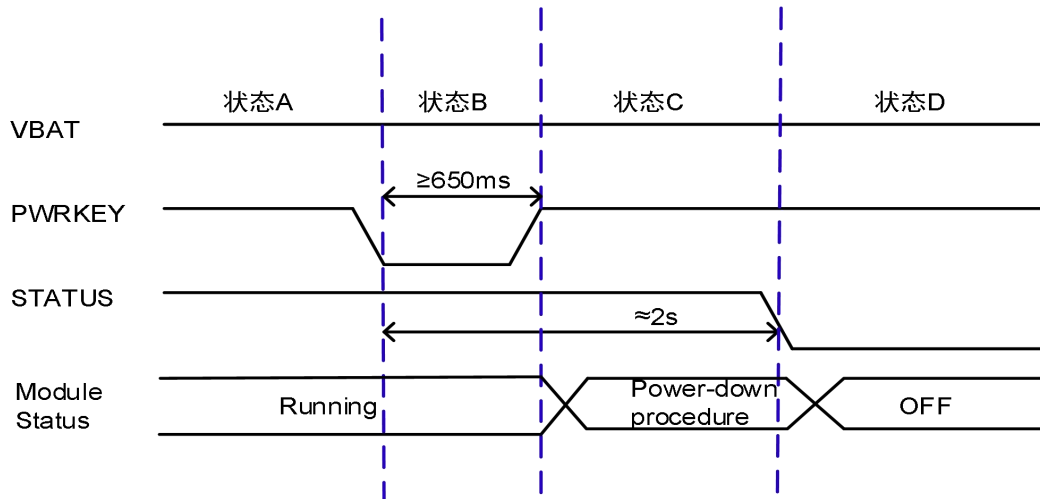


Figure 3.10 Power Key Shutdown Timing Diagram

PWRKEY shutdown timing description:

- ◆ Status A: Module working status;
- ◆ Status B: Lower the PERKEY status, duration at least 650ms or longer;
- ◆ Status C: The system inside the module shuts down, from pulling down PWRKEY to complete shutdown takes about 2 seconds;
- ◆ Status D: The module has entered the shutdown state.

3.6.2 Power off using AT command.

Executing AT+QPOWD or AT+LPOWD (please refer to the "AT Command Manual" for specific commands) can power off the module, and this operation has the same timing and effect as powering off by pulling down PWRKEY.

Notes

◆ When the module is operating normally, do not cut off the power supply immediately, otherwise it may damage the Flash data inside the module. It is recommended to first power off the module through PWRKEY or AT commands, and then disconnect the power supply.

◆ When shutting down using the AT command, please ensure that after the shutdown command is executed, the PWRKEY remains in a high level state. Otherwise, there may

be a situation where the device restarts after shutdown, which poses a risk of damaging the Flash if the power is disconnected at that time.

◆ If the device is powered on by grounding the PWRKEY, it is recommended to use AT+QPOWD/AT+LPOWD (selected according to the "AT Command Manual") to perform the shutdown process before disconnecting the power supply.

3.7 Reset

3.7.1 Reset Introduction

The RESET_N pin can be used for module reset. Pulling down the RESET_N pin for at least 300ms and then releasing it will reset the module. The description of the RESET_N pin is as follows:

Table 3-4 Reset Pin Description

Pin number	Pin names	I/O	Description	DC characteristics	Note
15	RESET_N	DI	Module reset	VILmax=0.5V	There is weak pull-up internally, low level effective, If not used, leave floating

3.7.2 Reset circuit reference design

The RESET_N signal is sensitive to interference. It is recommended that the wiring of the module interface should be as short as possible and grounded. The typical reference applications for hardware circuits are as follows, and the two circuit diagrams can be selected according to actual needs:

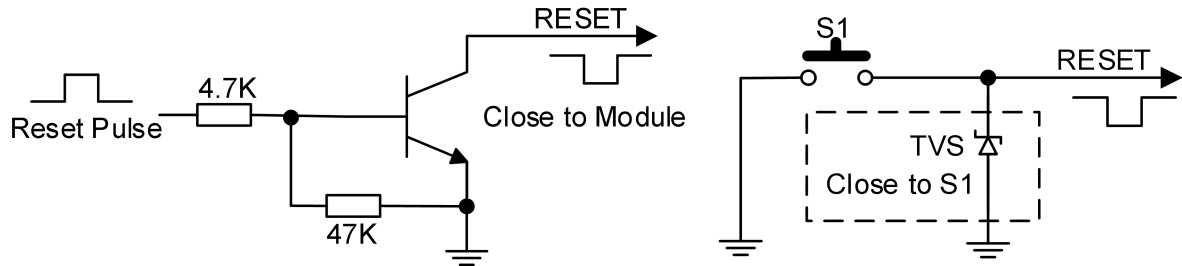


Figure 3.11 Reset Circuit Reference Design

3.7.3 Reset timing diagram

The reset timing sequence of the module is as follows:

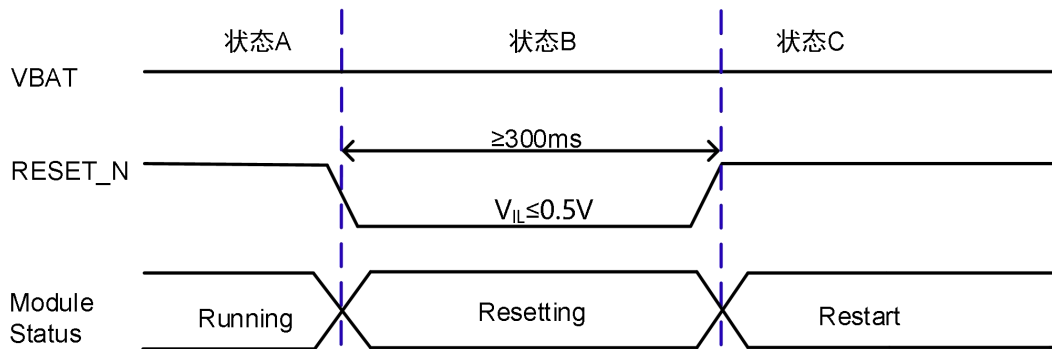


Figure 3.12 Reset Timing Diagram

Reset timing description:

- ◆ Status A: Module is in normal working condition;
- ◆ Status B: Pull down the state of RESET_N for at least 300ms;
- ◆ Status C: The module is starting to reboot.

4 Application Interface

The NT26KxBx series module provides rich peripheral functions, mainly including the following application interfaces:

- ◆ 1*USB2.0 interface*
- ◆ 3*UART interfaces
- ◆ 2*USIM interfaces
- ◆ 1*I2C interface
- ◆ 1*Simulated audio interface

4.1 USB interface*

The NT26KxBx series module supports USB 2.0 interface, complies with USB 2.0 specifications, and supports Full-Speed (12 Mbps) and High-Speed (480 Mbps) modes. The module only supports USB device mode. This interface can be used for AT command sending, data transmission, software debugging, and firmware upgrading. The interface definition is as follows:

Table 4-1 USB Interface Pin Description

Pin number	Pin names	I/O	Description	DC characteristics	Note
59	USB_DP	DIO	USB differential data (+)	-	Compliant with USB2.0 specifications. Requires 90 Ω differential impedance. Test points must be reserved. If not used, leave floating.
60	USB_DM	DIO	USB differential data (-)	-	Compliant with

					USB2.0 specifications. Requires 90 Ω differential impedance. Test points should be reserved. If not used, leave floating.
61	USB_VBUS	DI	USB inserted to wake up the module	$V_{max} = 5.25\text{ V}$ $V_{min} = 3.0\text{ V}$	When in sleep mode, inserting a USB will wake up the module.
82	USB_BOOT	DI	USB mode configuration interface	VO_LDOIO	Internal weak pull-down, high level effective. Before the module starts up normally, it is forbidden to pull up this pin. It is recommended to reserve a test point. If not used, leave floating.

4.1.1 USB circuit reference design

The module's USB interface can be connected to a USB connector or the MCU's USB interface. The reference schematic for connecting to the USB connector is as follows:

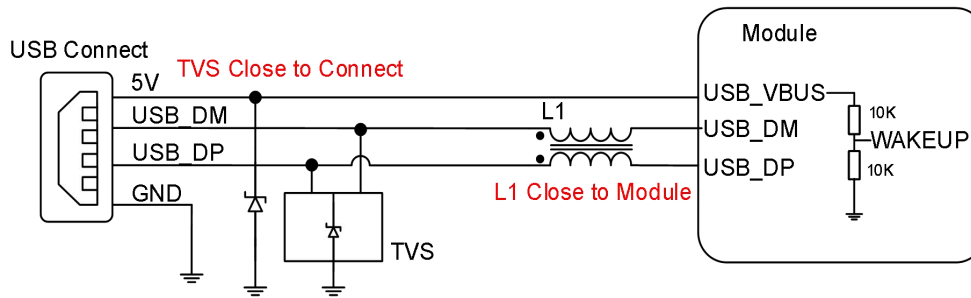


Figure 4.1 USB Interface Reference Design

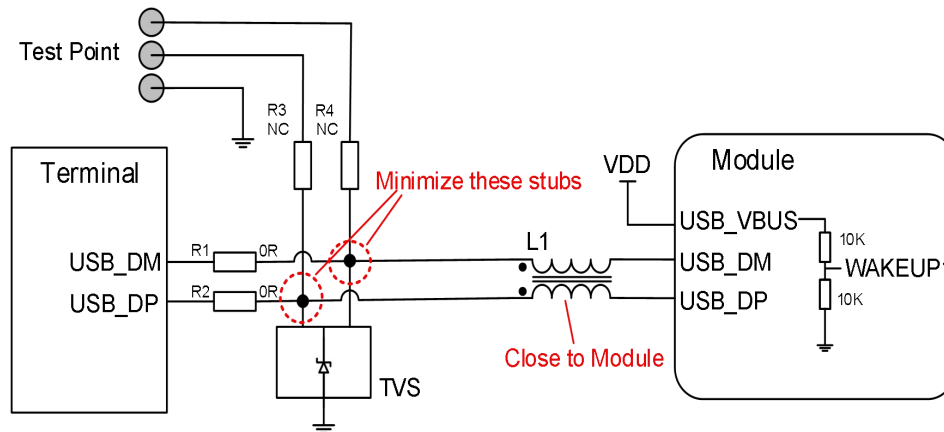


Figure 4.2 USB connection MCU reference design

Notes

◆ To ensure the performance of USB, the routing of USB_DP and USB_DM should maintain a differential impedance of $90\ \Omega$, and the surrounding area needs to be treated with a ground connection. It is recommended to place a common mode inductor L1 between the MCU and the module to prevent USB signal from generating EMI interference, and L1 should be placed close to the module.

◆ USB traces should be kept away from crystal oscillators, oscillators, magnetic devices, RF signals, and strong signal areas. It is recommended to use inner layer differential traces and surround them with GND on all sides.

If the module's USB interface and MCU are connected via a connector, a TVS diode protection should be added at the interface. Special attention should be paid to the selection of ESD protection devices, ensuring that the parasitic capacitance does

not exceed 2pF, and they should be placed as close to the USB interface as possible.

◆ Reserve resistors R1~R4, normally connect MCU by attaching R1~R2, do not attach R3~R4; when debugging is needed, attach R3~R4, do not attach R1~R2 to switch to the test point. In order to meet the USB data line signal integrity requirements, the bifurcation lines connecting the test points should be kept as short as possible.

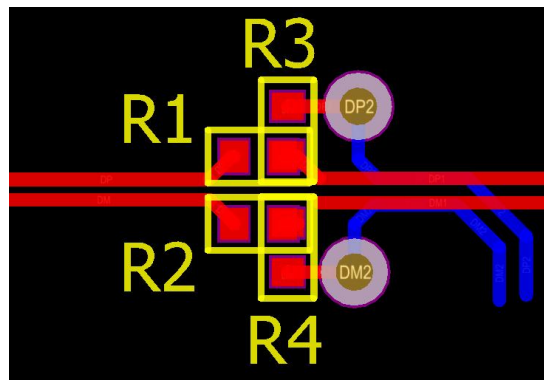


Figure 4.3 USB jumper routing diagram

4.1.2 USB data transfer

When the module is powered on or reset, if USB_BOOT is detected as a low level, the USB port will enumerate 2 serial ports for AT command transmission and log output.

4.1.3 USB firmware download

When the module is powered on or reset, if USB_BOOT is detected as a high level, a dedicated download serial port will be enumerated on the USB port, and the download tool can perform firmware upgrades through the USB interface. If no download tool is connected for more than 15 seconds, the module will re-enumerate 2 serial ports according to the default USB configuration.

The reference design for the USB_BOOT interface is as follows:

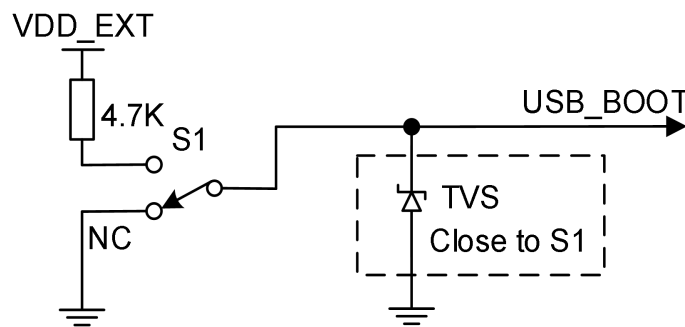


Figure 4.4 USB_BOOT Reference Design

4.2 UART communication

The module provides 3 general asynchronous transceivers: main serial port, debug serial port, and auxiliary serial port. The baud rate of the main serial port is configurable, and the debug serial port is only used for debugging and testing. The pin definitions for the serial ports are as follows:

Table 4-2 Serial Port Pin Definitions

Interface	Pin number	Pin names	Describe	Note
Main serial port	17	MAIN_RXD	Main serial port receives data	Non-sleep mode: VDD_EXT power domain; Sleep status: switch to internal AON power domain
	18	MAIN_TXD	Main serial port sends data	If not used, leave floating
	19	MAIN_DTR	Master serial port data terminal ready	In sleep mode, the wake-up pin can be used.
	20	MAIN_RI	Main serial port outputs ringing	If not used, leave floating

			prompt.	
	21	MAIN_DCD*	Main serial port output carrier detection	Not supported at the moment.
	22	MAIN_CTS*	Main serial port DTE clear to send	Internally connected in parallel with pin 57, cannot be used simultaneously.
	23	MAIN_RTS*	The main serial port DTE requests to send.	Internally connected in parallel with pin 58, cannot be used simultaneously.
Debug serial port	38	DBG_RXD	Debug serial port to receive data	If not used, leave floating
	39	DBG_TXD	Debug serial port to send data	If not used, leave floating
Auxiliary serial port	28	AUX_RXD*	Assist in receiving data through the serial port.	Not supported at the moment
	29	AUX_TXD*	Assist in sending data through the serial port.	Not supported at the moment.

Precautions

◆ The serial port interface belongs to the V0_LD0I0 power domain, with a default voltage level of 1.8V;

◆ In sleep mode, the V0_LD0I0 power domain will lose power, and MAIN_RXD will switch to the internal 1.2V power supply.

◆ The MAIN_RXD pin has internal pull-up, no external pull-up resistor is needed, and it has anti-reverse flow function.

◆ Pay attention to the consistency of the voltage levels when using the serial port, otherwise it may cause leakage current.

4.2.1 Serial port application

Main serial port characteristics:

- ◆ Used for AT command communication and data transmission, default baud rate is 115200bps (8-N-1);

- ◆ Configurable as:
4800/9600/19200/38400/57600/115200/230400/460800/921600bps.

- ◆ Used for firmware upgrade, the default baud rate during the upgrade is 921600bps;

- ◆ Support low power consumption wake-up module function.

Characteristics of serial port debugging:

- ◆ Through the dedicated tools provided by the platform, you can view log information for software debugging, with a default speed of 3000000bps.

Serial port connection diagram is as follows:

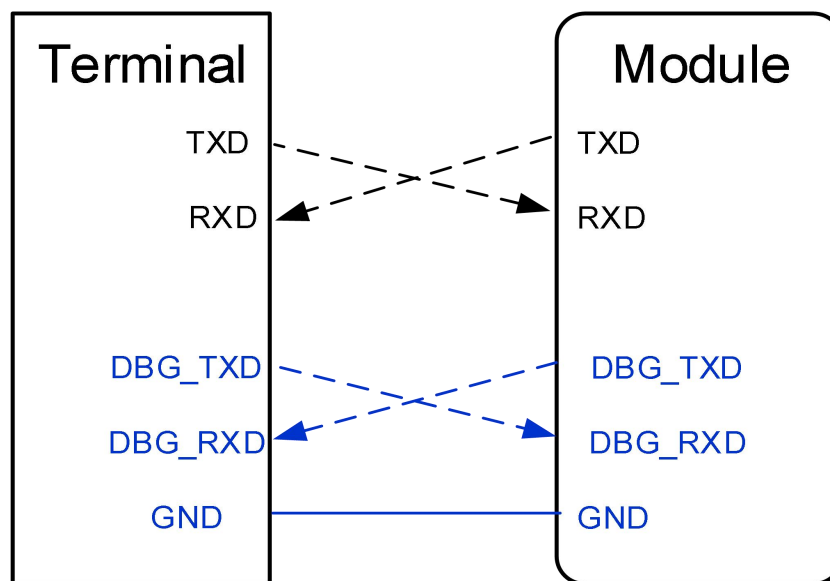


Figure 4.5 Serial Port Connection Diagram

4.2.2 Serial port circuit reference design

The main considerations for a suitable serial port level conversion circuit include whether it meets the working speed of the serial port, scenarios with low power

requirements, and whether the power consumption meets the requirements, etc. It is recommended to refer to the following level matching circuit:

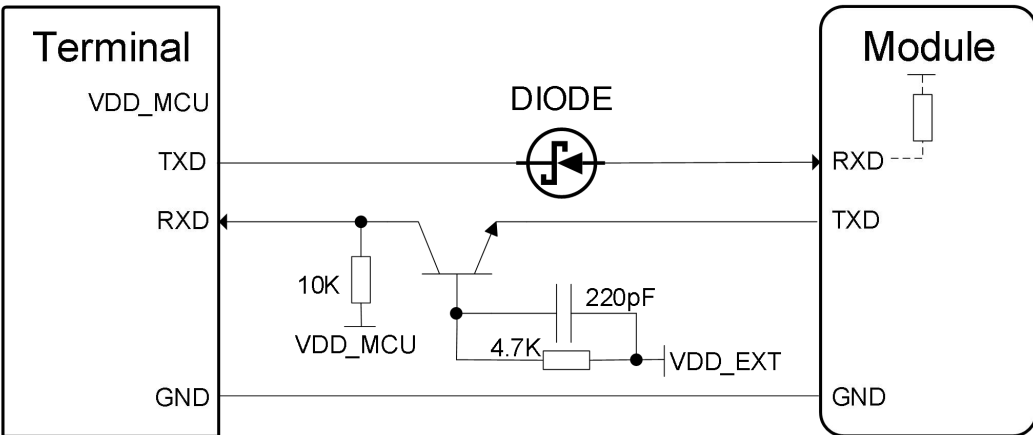


Figure 4.6 Transistor Level Conversion Reference Circuit

Recommend three transistors for reference:

Brand: CJ Model: S8050 J3Y Package: SOT-23

Recommended diode for reference (note that the forward voltage of the diode should be $\leq 0.3V$):

Brand: LRC Model: LRB520S-30T1G Package: SOD-523

4.3 USIM card interface

The module includes 2 external USIM card interfaces. Card 1 complies with ETSI and IMT-2000 specifications, supporting 1.8V and 3.0V USIM cards, while Card 2 defaults to supporting 1.8V USIM cards. The pin descriptions are as follows:

Table 4-3 Definition of External USIM Card Interface Pins

Pin number	Pin names	Description	DC characteristics
11	USIM_DATA	SIM card data cable	VO_LDOSIM
12	USIM_RST	SIM card reset line	VO_LDOSIM
13	USIM_CLK	SIM card clock line	VO_LDOSIM

15	USIM_VDD	SIM card power supply	Vnorm=1.8/3.0V
79	USIM_DET	SIM card status signal detection	SIM card hot plug detection interface
62	USIM2_CLK	SIM2 (sub card) clock line	VO_LDOIO
63	USIM2_RST	SIM2 (secondary card) reset line	Internally connected in parallel with pin 50, cannot be used simultaneously.
64	USIM2_DATA	SIM2 (secondary card) data cable	Internally connected in parallel with pin 51, cannot be used simultaneously.
65	USIM2_VDD	SIM2 (secondary card) power supply,	Powered by the same source as USIM_VDD.

The module supports hot-swappable USIM cards through USIM1_DET, and also supports high/low level detection simultaneously. This function is disabled by default and can be configured via AT+QSIMDET. If the USIM card detection function is not needed, USIM_DET can be left floating.

4.3.1 USIM card circuit reference design

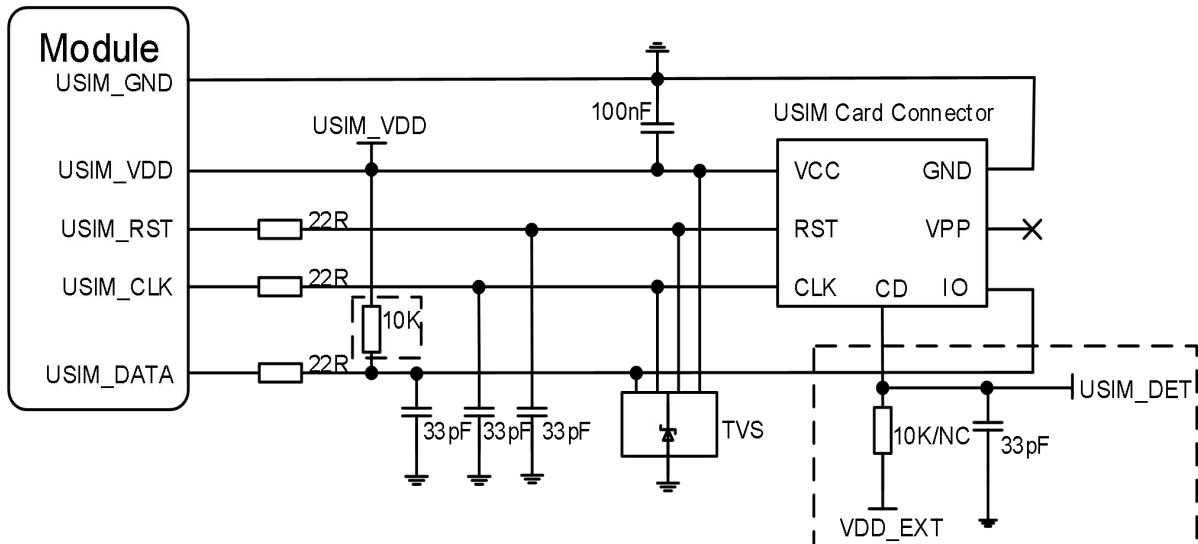


Figure 4.7 6PIN SIM card reference circuit

Precautions

To ensure that the power consumption in the sleep state meets the design standards, it is recommended to use a SIM card slot with CD foot normally closed.

The hardware design circuit of USIM2 can also refer to the above design, specifically refer to the "Lierda NT26-FEU D Series Hardware Reference Design Manual".

4.3.2 Considerations for USIM card circuit design

To ensure the reliability and availability of the SIM card in the application, please follow the following USIM circuit design guidelines:

- ◆ When laying out, try to place the SIM card as close to the module as possible, and keep the signal wire layout length within 200mm if possible.

- ◆ SIM card signal line away from RF trace and VBAT power line;

- ◆ The GND wiring of the SIM card should be short and thick, ensuring that the wiring width is not less than 0.5mm;

- ◆ The decoupling capacitance of USIM_VDD should not exceed 1uF, and the capacitance should be placed close to the external SIM card's VCC.

- ◆ To avoid signal interference between USIM_DATA and USIM_CLK, the wiring between them should not be too close. Ground shielding should be added between the two traces. In order to prevent the negative impact of long traces, a pull-up resistor is generally added to USIM_DATA to USIM_VDD to enhance driving capability. If the traces are too long, it is recommended to externally reserve a 10K resistor near the card slot. In addition, the USIM_RST signal also needs to be protected by ground.

To ensure good ESD protection performance, it is recommended to add a TVS diode to the pins of the external USIM card. The parasitic capacitance of the selected TVS diode should not exceed 15pF; a 22 ohm resistor should be connected in series between the signal lines of the module and the SIM card to suppress stray EMI and enhance ESD

protection. Additionally, a parallel 33pF capacitor should be used to filter out RF interference. The relevant resistors, capacitors, and TVS diode should be placed near the USIM card slot.

4.4 I2C interface

The module provides 1 I2C interface, and the interface description is as follows:

Table 4-4 I2C Interface Pin Definitions

Pin number	Pin names	Description	DC characteristics	Note
66	I2C_SDA	I2C data	VO_LDOIO	I2C data signal, need to add an external 4.7K pull-up resistor, leave floating if not used
67	I2C_SCL	I2C clock	VO_LDOIO	I2C clock signal, need to add an external 4.7K pull-up resistor, leave floating if not used.

4.5 Simulated audio interface

4.5.1 Simulated audio introduction

Table 4-5 SPK Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
5	SPK_P	Audio differential output	VO_LDOIO	Connected in parallel with pin 82 internally, cannot be used simultaneously.
6	SPK_N	Audio negative differential output	GND	If not used, leave floating

Note

Audio output channels can be used as speakers (if the output power is insufficient, it is recommended to add an external audio amplifier component), receiver output channels, for increased anti-interference capability, it is recommended to use differential output.

4.5.2 Simulated audio reference application

By using an external audio PA, high-power speaker playback can be achieved, a common reference circuit is as follows:

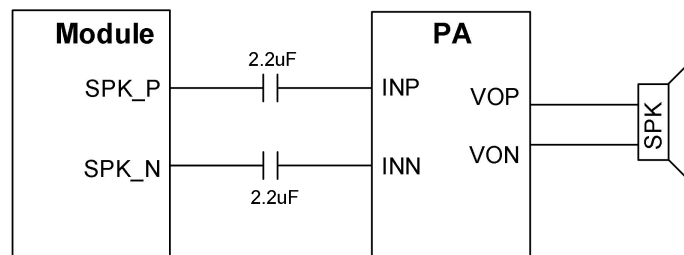


Figure 4.8 SPK Application Reference Circuit

In actual applications, the selected PA will determine the circuit design accordingly. For specific circuit details, please refer to the "Lierda NT26KxBx Series Hardware Reference Design Manual" document.

4.5.3 Audio application notes

The severity of high-frequency interference during GSM transmission usually depends on the application design. In some cases, TDD noise under EGSM900 is more severe, while in other cases, TDD noise under DCS1800 is more severe. Therefore, the required filtering capacitors can be selected based on actual test results. The placement of RF filtering capacitors on the PCB should be as close as possible to the audio device or audio interface, with routing kept as short as possible, passing through the filtering capacitors before reaching other points.

To reduce radiation interference, the antenna should be positioned as far away as possible from audio components and audio wiring; power lines and audio wiring should not

run parallel to each other, and power lines should be kept as far away as possible from audio wiring.

4.6 ADC interface

The module provides 2 channels of 12-bit analog-to-digital conversion input interfaces (ADC interfaces). When wiring the ADC interface, in order to improve the voltage measurement accuracy of the interface, it is recommended to perform grounding treatment.

The pin definitions are as follows:

Table 4-6 ADC Pin Definitions

Pin number	Pin Name	Description	DC characteristics	Note
9	ADC0	General ADC interface	Input voltage range: 0~1.05V	If not used, leave floating
96	ADC1	Universal ADC interface		If not used, leave floating

Using AT+QADC=<port> can read the voltage values of each pin of the ADC interface.

- ◆ AT+QADC=0 or AT+LADC=0 (please refer to the "AT Command Manual" for specific commands): can be used to read the voltage value of ADC0
- ◆ AT+QADC=1 or AT+LADC=1 (For specific commands, please refer to the "AT Command Manual"): can be used to read the voltage value of ADC1

Table 4-7 ADC Characteristics

Parameters	Minimum value	Typical value	Maximum value	Note
ADC voltage range	0	-	1.05	V
ADC input impedance	0.26	-	0.75	Ω
ADC accuracy	-	12	-	bits

Notes

If the collected voltage is greater than 1.05V, it is recommended to use a resistor divider circuit for the input of the ADC pin. The resistance value of the divider resistor should not exceed 100K Ω , and it is recommended to use resistors with an accuracy of 1% or higher, otherwise it will reduce the measurement accuracy of the ADC.

◆ When designing, a 1nF capacitor can be optionally placed across the two ends of the ground potential divider resistor.

4.7 Status indicator signal

The module has 1 network status indicator and 1 running status indicator interface, and the interface is defined as follows:

Table 4-8 Status Indicator Pin Definitions

Pin number	Pin name	Description	DC characteristics		Note
16	NET_STATUS	Network status indicator	VDD18AON	If not used, leave floating	
25	STATUS	Operating status indicator	LDO_AONIO/ VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.	

4.7.1 Network status indicator

The NET_STATUS pin is used to indicate the network status of the module, and the logic levels under different statuses are as follows:

Table 4-9 Definition of Status Indicator Pins

Pin name	Level logic	Network status
NET_STATUS	Slow flash (200ms high/1800ms low)	Search network status
	Slow Flash (1800ms high/200ms low)	Standby mode
	Quick Flash (125ms high/125ms low)	Data transmission mode

4.7.2 Running status indicator

The STATUS pin is used to indicate the operating status of the module. When the module is powered on normally, STATUS will output a high level, and it remains valid even in sleep mode.

4.7.3 Indicator light circuit reference design

NET_STATUS and STATUS pins can both be used to drive LED indicator lights, the reference design is as follows:

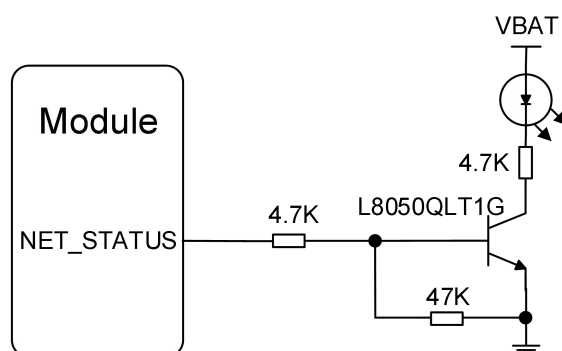


Figure 4-9 Network Status Indicator Light Reference Circuit

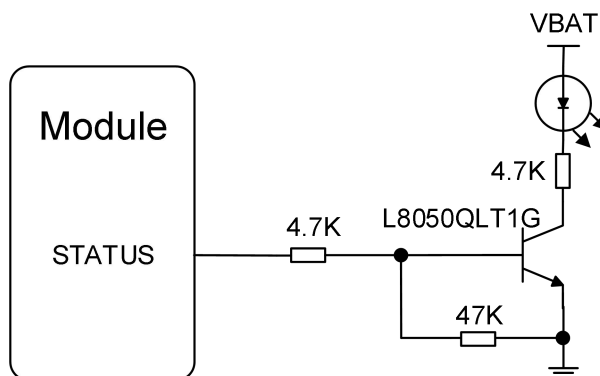


Figure 4-10 Module Status Indicator Light Reference Circuit

5 Radio frequency characteristics

5.1 Introduction to RF Function

The NT26KxBx series module provides a single RF interface with an impedance of 50 Ω . The interface description is as follows:

Table 5-1 RF Interface Pin Definitions

Pin number	Pin name	I/O	Description	Note
35	ANT_MAIN	AIO	Main antenna interface	50 Ω characteristic impedance

5.2 Cellular network

5.2.1 Working frequency band

The operating frequency bands supported by the NT26KxBx series module are as follows:

Table 5-2 Working Frequency Bands

Working frequency band	Transmit (MHz)	Receive (MHz)
LTE-FDD B1	1920~1980	2110~2170
LTE-FDD B3	1710~1785	1805~1880
LTE-FDD B5	824~849	869~894
LTE-FDD B8	880~915	925~960
LTE-TDD B34	2010~2025	2010~2025
LTE-TDD B38	2570~2620	2570~2620
LTE-TDD B39	1880~1920	1880~1920
LTE-TDD B40	2300~2400	2300~2400

LTE-TDD B41	2535~2675	2535~2675
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Notes

LTE-TDD B41 only supports 140M (2535-2675MHz).

5.2.2 Transmit power

The transmission power of the NT26KxBx series module is as follows:

Table 5-3 Conduction Power

Frequency band	Maximum value	Minimum value	Note
LTE-FDD B1 (10 MHz)	23dBm±2.7dB	<-40dBm	Compliant with Cat.1 protocol in 3GPP Rel-13 and Rel-14.
LTE-FDD B3 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-FDD B5 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-FDD B8 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B34 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B38 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B39(10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B40 (10 MHz)	23dBm±2.7dB	<-40dBm	
LTE-TDD B41 (10 MHz)	23dBm±2.7dB	<-40dBm	

5.2.3 Sensitivity of reception

The receiving sensitivity of the NT26KxBx series modules is as follows:

Table 5-4 Conduction reception sensitivity under single transmission (throughput ≥95%)

Frequency band	Receive sensitivity (10M)	3GPP standard (10M)
LTE-FDD B1 (10 MHz)	-97.8dBm	-96.3dBm
LTE-FDD B3 (10 MHz)	-99.8Bm	-93.3dBm
LTE-FDD B5 (10 MHz)	-98.3dBm	-94.3dBm
LTE-FDD B8 (10 MHz)	-98.9dBm	-93.3dBm

LTE-TDD B34 (10 MHz)	-101dBm	-96.3dBm
LTE-TDD B38 (10 MHz)	-100.4dBm	-96.3dBm
LTE-TDD B39(10 MHz)	-101.3dBm	-96.3dBm
LTE-TDD B40 (10 MHz)	-100.8dBm	-96.3dBm
LTE-TDD B41 (10 MHz)	-100.3dBm	-94.3dBm

5.3 Antenna reference circuit design

When using the NT26KxBx series module, a π -type matching circuit needs to be added between the RF antenna interface of the module and the antenna interface of the user's baseboard. The typical antenna matching circuit and initial parameters are shown in the figure below:

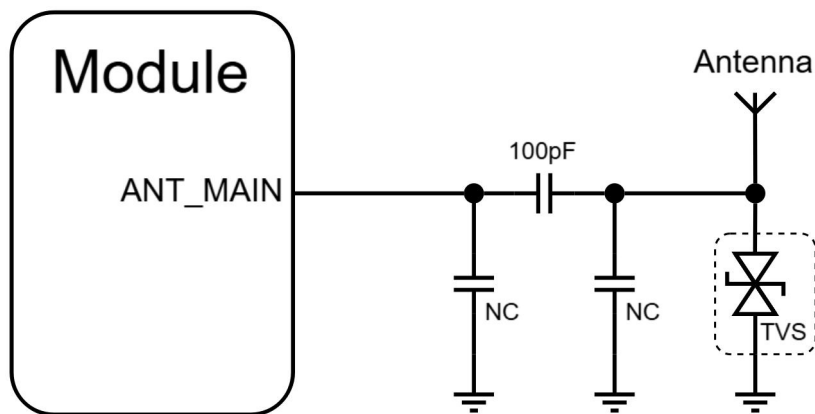


Figure 5.1 RF Antenna Reference Circuit

Notes

◆ The series capacitance is 100pF, the capacitor position is default unpopulated, and the recommended package for the component is 0201 or 0402.

◆ When using an external antenna, or when users can touch the antenna, it is recommended to reserve a TVS diode for enhanced electrostatic protection. Due to the parasitic capacitance of the TVS diode which may affect the antenna performance, a model

with parasitic capacitance $\leq 0.5\text{pF}$ should be selected, and it is recommended to re-calibrate the antenna after adding the TVS diode.

5.4 RF signal line wiring guidance

The routing between the module antenna interface and the user antenna must meet the 50-ohm characteristic impedance requirements. At the same time, the RF routing should be as short as possible to ensure minimal insertion loss of the RF routing.

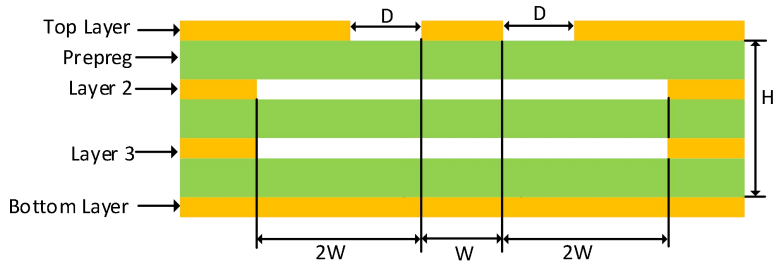
In general, the impedance of radio frequency signal lines is determined by the material's dielectric constant ϵ_r , trace width W , spacing to ground D , and the thickness of the reference ground plane H .

In the field of IoT applications, the design of PCB characteristic impedance typically adopts a coplanar waveguide approach to achieve better shielding for RF signal lines, while also achieving higher integration for compact design.

Common PCB coplanar waveguide designs include the following:

Table 5-5 Description of Common PCB Coplanar Waveguide Structures

Serial Numb er	Pin names	Note
1	Top Layer → Prepreg → Bottom Layer → Dimensions: D (gap), W (trace width), H (prepreg thickness).	2-layer board Reference ground is L2 layer.
2	Top Layer → Prepreg → Layer 2 → Layer 3 → Bottom Layer → Dimensions: D (gap), W (trace width), H (prepreg thickness), $2W$ (bottom layer side sections), W (bottom layer central gap).	4-layer board Reference ground is at level L3 L2 layer excavation

3		4-layer board Reference ground is L4 layer. Digging out at levels 2 and 3.
---	--	---

During the layout process, 50 ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on actual conditions, appropriate line width, line spacing, and stack-up should be selected to achieve the best design effect.

Take the example of a 2-layer PCB with a thickness of 1.6mm coplanar waveguide for calculation.

◆ Choose the appropriate PCB substrate material, once the material is selected, the dielectric constant ϵ_r is basically determined (in this case, it is 4.6);

◆ To choose the appropriate line width, a good method is to select a line width that matches the size of the pads left by the device (note that it should not be too thin, choose 0.65mm in this example);

- According to the selected board thickness, line width, dielectric constant, etc., the approximate distance to GND can be calculated. If the line spacing is not suitable, the line width can be adjusted to adjust the line spacing (calculated to be 0.14mm, which is basically suitable).

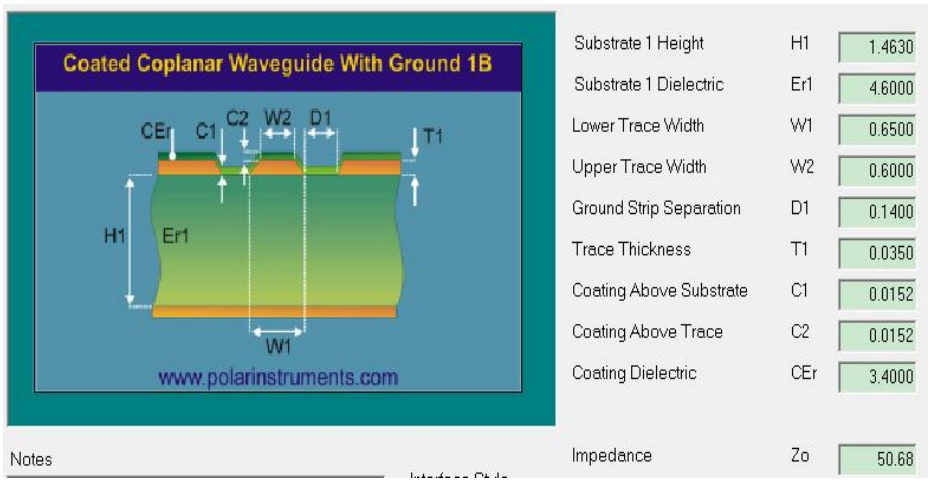


Figure 5.2 Reference for Calculation of 50 Ohm Impedance

During the actual layout process, the following suggestions are provided for reference:

- ◆ In the π -type circuit, three matching reserved components are placed closely near the antenna, while bypass components are recommended to be placed on both sides of the RF line.
- ◆ Irregular vias are often placed on both sides of the RF traces, and there must be a complete GND plane below the entire RF trace space.
- ◆ There should be no other traces below the RF line to avoid affecting the RF performance or other circuits.

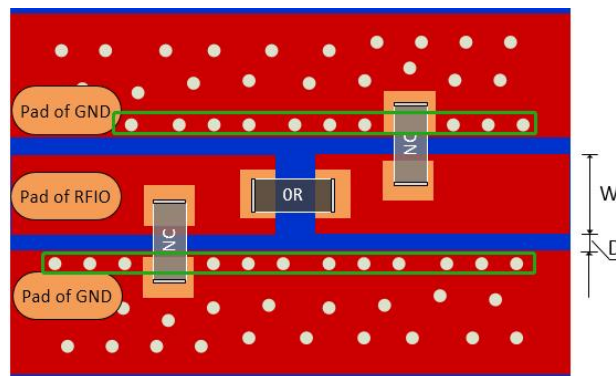


Figure 5-3 RF Routing Layout Design Schematic

The reasonableness of RF routing can be directly reflected in the module's conducted tests. In order to ensure the full performance of the product, cooperation with antenna design is also required. To better meet the requirements of antenna design, the following requirements are hoped to be achieved in PCB design. Guidance will be provided for the whole machine PCB at different layers:

- When the product PCB is designed with 2 layers, it is best for both the TOP and BOTTOM LAYER directly below the module to be GND layers. The traces that need to be led out from the module should avoid passing directly below the module and should be led out from the outer side of the module.

- ◆ When the product PCB is designed with 4 layers, it is recommended that the traces to be led out by the module should be routed on the third or fourth layer, and the first and

second layers should be reserved as complete GND reference layers for the module.

5.5 Antenna design requirements

The RF performance of the module is also affected by the antenna. The selection of the antenna needs to meet the following requirements:

- ◆ Select an antenna that matches the operating frequency band of the module.
- ◆ The characteristic impedance of the antenna is required to be 50 ohms to reduce the loss at the connection between the RF cable and the antenna.
- ◆ The lower the insertion loss within the working frequency band, the better, such as $VSWR \leq 2$, return loss $\geq 10\text{dB}$, etc.;

Commonly used antennas suitable for CAT1 scenarios are as shown in the following figure:

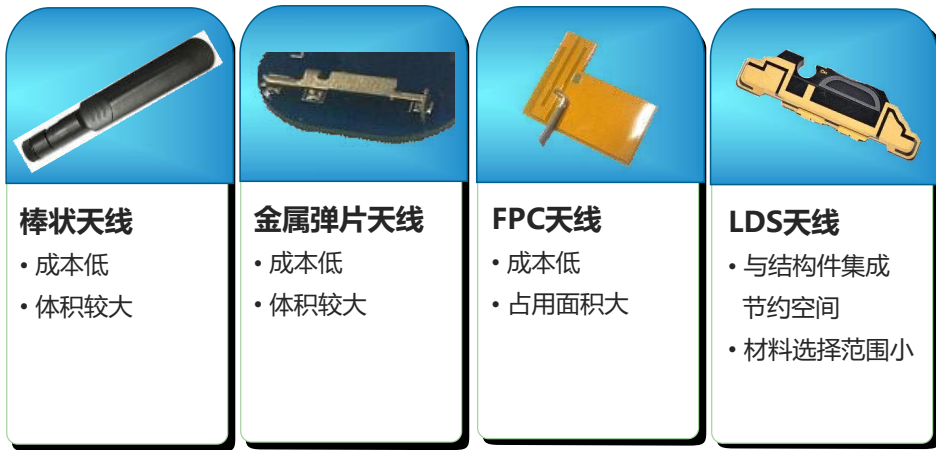


Figure 5.4 Commonly Used Antenna Types for CAT1

5.6 Recommended RF connectors

In some applications where customized antennas are needed, such as in environments where devices are shielded by metal casings, IPX or SMA connectors can be left on the motherboard to connect the antenna to the outside of the device through a coaxial cable to achieve better performance.

Table 5.6 Common RF Connectors

Serial number	Product name	Image	Note
1	IPX socket		Connect the RF cable.
2	SMA constellation		Can connect to RF cable or antenna.

6 Electrical performance and reliability

6.1 Absolute maximum rating

The description of the absolute maximum rated value of the module interface is as follows:

Table 6-1 Absolute Maximum Ratings

Parameters	Minimum value	Maximum value	Unit
VBAT	-0.3	5.0	V
USB_VBUS voltage	-0.3	5.25	V
Digital interface voltage	-0.3	3.6	V
VBAT current	-	1.2	A

Notes

When the operating conditions exceed the absolute maximum ratings, it may cause permanent damage to the module.

6.2 Rated power value

The electrical characteristics of the module power interface are described as follows:

Table 6-2 Power Interface Electrical Characteristics

Parameter s	Description	Conditions	Minim um value	Typica l value	Maxi mum value	Unit
VBAT	Module normal supply voltage(1)	The actual input voltage must be within this range.	3.3	3.8	4.5	V
VBAT	Module expansion power supply voltage	The actual input	3.1	3.8	4.5	V

	(2)	voltage must be within this range.				
I_{VBAT}	Peak current (each Transmission time slot)	Under the maximum transmit power level	-	1.0	1.2	A
USB_VBUS	USB detection	-	3.0	5.0	5.25	V
USIM_VDD	USIM card power supply output	-	-	1.8/3.0	-	V
VDD_EXT	Reference power output	-	-	1.8	-	V

Notes

◆ When the module operates within this voltage range, the module's relevant performance meets the 3GPP standard requirements.

- When the module operates within this voltage range, it can still maintain normal working status without any irreparable faults; only individual indicators, such as output power and other parameters, may exceed the range specified in the 3GPP standard. When the voltage returns to the normal operating range, all the module's indicators still comply with the 3GPP standard.

◆ Sleep Mode: VDD_EXT does not power off in sleep mode.

6.3 Power consumption

The power consumption of the module is as follows:

Table 6-3 Module Power Consumption

Module description	Test conditions	Typical value	Unit
Shutdown mode	Module shutdown	0.74	uA

Sleep Mode	AT+CFUN=0 (USB disconnected)	6.5	uA
	LTE-FDD @ PF = 32 (USB disconnected)	0.78	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.41	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.22	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.14	mA
	Lierda LTE-TDD @ PF = 32 (USB disconnected)	0.75	mA
	Lierda LTE-TDD @ PF = 64 (USB disconnected)	0.39	mA
	LTE-TDD @ PF = 128 (USB disconnected)	0.21	mA
	LTE-TDD @ PF = 256 (USB disconnected)	0.15	mA
Idle mode	LTE-FDD @ PF = 64 (USB disconnected)	2.3	mA
	LTE-FDD @ PF = 64 (USB connection)	22.4	mA
	Lierda LTE-TDD @ PF = 64 (USB disconnected)	2.9	mA
	LTE-TDD @ PF = 64 (USB connection)	22.2	mA
LTE data transmission	LTE-FDD B1	617	mA
	LTE-FDD B3	590	mA
	LTE-FDD B5	592	mA
	LTE-FDD B8	593	mA
	LTE-TDD B34	225	mA
	LTE-TDD B38	222	mA
	LTE-TDD B39	205	mA
	LTE-TDD B40	239	mA
	LTE-TDD B41	223	mA

6.4 Digital logic voltage characteristics

The module defines the logic levels of digital IO as follows:

Table 6-4 Explanation of Digital IO Logic Levels

Type	Parameter	Description	Minimum	Maximum	Unit
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	s		value	value	
Input	V _{IL}	Low level input	-0.3	0.6	V
	V _{IH}	High level	1.2	2	V
Output	V _{OL}	Low level output	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Notes

The GPIO reference level of the module follows VDD_EXT, and VDD_EXT will lose power during sleep, so do the corresponding GPIOs.

Power off.

The AGPIO logic level definition for the NT26-FEU D module is as shown in the table below: (AON_VDD range: 1.7~1.9V; typical value: 1.8V)

Table 6-5 Explanation of Digital IO Logic Levels

Type	Parameter s	Description	Minimum value	Maximum value	Unit
Input	V _{IL}	Low level input	-0.3	0.6	V
	V _{IH}	High level	1.2	2	V
Output	V _{OL}	Low level output	-	0.45	V
	V _{OH}	Output high level	1.35	-	V

Notes

The AGPIO reference level of the module follows LDO_AONIO and will not lose power during sleep.

The logic voltage levels of the USIM card interface module are defined in the following

table:

Table 6-6 Explanation of USIM 1.8V Logic Level

Type	Parameters	Describe	Minimum value	Maximum value	Unit
Input	V_{IL}	Low level input	-	0.6	V
	V_{IH}	High level	1.2	-	V
Output	V_{OL}	Low level output	-	0.45	V
	V_{OH}	Output high level	1.35	-	V

Table 6-7 Explanation of USIM 3.0V Logic Level

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	V_{IL}	Low level input	-	1.0	V
	V_{IH}	High level	1.95	-	V
Output	V_{OL}	Low level output	-	0.45	V
	V_{OH}	Output high level	2.55	-	V

Precautions

The voltage level of USIM_VDD will be determined based on the detected category of the USIM card, supporting 1.8/3.0V USIM cards.

6.5 Static electricity protection

In life and production, static electricity is everywhere. Static electricity generated by human body static electricity, object friction, etc., can be transferred to the module through various means, and may cause damage to the module. Therefore, it is necessary to pay close attention to static electricity protection and take necessary static electricity protection measures. For example:

◆ Wear anti-static gloves during the research and development, production, assembly, and testing processes;

◆ When designing products, add electrostatic protection devices at circuit interfaces and other points susceptible to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-8 Electrostatic Protection Characteristics

Test interface	Discharge phenomenon	Note
VBAT	$\pm 8\text{KV}$	Test standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	$\pm 8\text{KV}$	
Antenna interface	$\pm 8\text{KV}$	
Others	-	

Electrostatic protection detailed design refers to "Lierda NT26KxBx Series Hardware Reference Design Manual".

6.6 Working and storage temperature

Table 6-9 Working Temperature Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Note
Normal operating temperature ⁽¹⁾	-35	+25	+75	°C	
Expand operating temperature range ⁽²⁾	-40	+25	+85	°C	
Storage environment temperature ⁽³⁾	-40	+25	+90	°C	

Notes

(1) When the module operates within this temperature range, the module's relevant performance meets the 3GPP standard requirements.

(2) When the module operates within this temperature range, the module can still

maintain normal operation without irreversible failures; only individual indicators, such as output power and other parameters, may exceed the range specified in the 3GPP standard. When the temperature returns to the normal operating range, all indicators of the module still comply with the 3GPP standard.

(3) This storage temperature range, excluding packaging materials, needs to pay attention to the highest temperature tolerance of the tape packaging.

6.7 Notes

Modules need attention in the production and usage process:

- ◆ When spraying the module, please try to avoid the spraying material from flowing into the interior of the module.

- When cleaning the module, do not use ultrasonic cleaning on the module, as it may cause damage to the internal crystals of the module.

- ◆ When producing and using modules, avoid applying them in environments or enclosures containing any amount of mercury or mercury vapor, as this may lead to product failure or malfunction risks.

7 Mechanical dimensions

7.1 Mechanical dimensions

The mechanical dimensions diagram of the module is as follows:

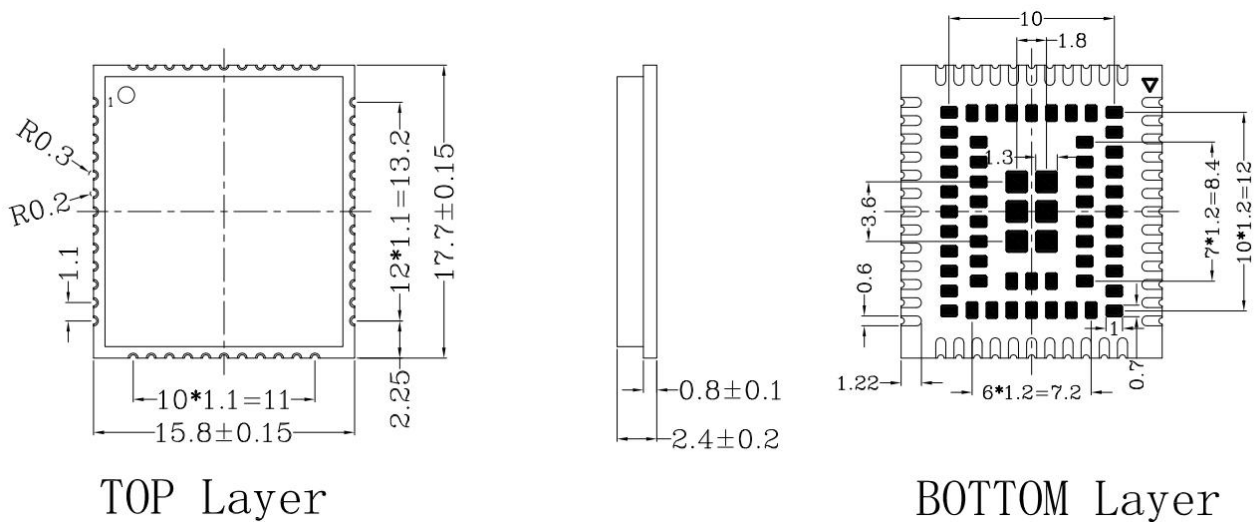


Figure 7.1 Module Mechanical Dimensional Drawing (unit: mm)

7.2 Module top view/bottom view

The physical effect diagram of the module is as follows, please refer to the actual module for details:



Figure 7.2 Module Top/Bottom View

7.3 Recommend encapsulation

The module recommends solder pads as shown in the following figure, and users can make minor adjustments according to their own production processes.

◆ The pins around the module are designed with rounded corners inside. When designing the bottom board for soldering, please consider using rounded corners for transition. The rectangular solder pads at the bottom of the module can be designed with the module pin dimensions when designing the bottom board, as shown in the reference design diagram below.

◆ In order to facilitate the opening of the stepped steel mesh, it is recommended not to layout other components within 2.0mm range outside the module solder pad. Users can refer to the requirements of their own steel mesh manufacturers to determine this distance.

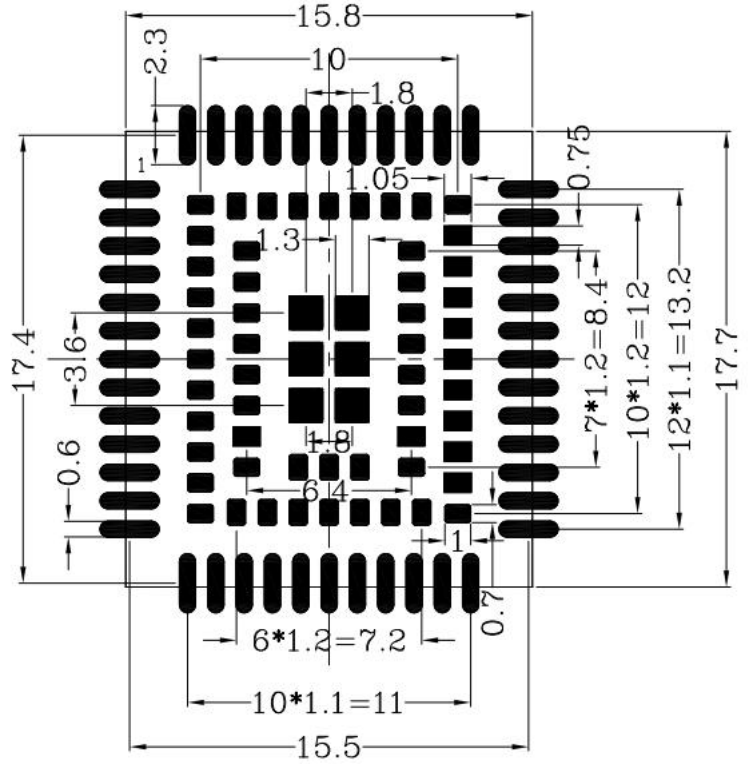


Figure 7.3 Module Recommended Solder Pads (unit: mm)

8 Production and packaging information

This chapter describes information such as the patch process, packaging method, storage conditions of the module, which can help users better store and use this module.

8.1 Production welding

8.1.1 Furnace passing method

When the module is being mounted, the following points need to be noted during reflow soldering:

- ◆ To ensure the quality of module soldering, it is recommended to use a steel mesh thickness of 0.18-0.2mm for the module solder pad area.
- ◆ If the bottom of the module is double-sided patch, it is recommended to place the module in the second patch to avoid damage to the module due to repeated heating;
- ◆ It is best for the customer's substrate to go through the reflow oven on the mesh belt for the first SMT placement, and for the second SMT placement, try to put it on the mesh belt for reflow as well.
- ◆ If for special reasons it is not possible to put it on the mesh belt for reflow soldering, consider using fixtures for reflow soldering on the track or placing a flat, high-temperature resistant straight template under the PCBA to prevent PCB deformation during reflow soldering, leading to virtual welding of the module.

8.1.2 Soldering operation guidance

PCBA reflow soldering temperature curve is related to the use of solder paste, and needs to be adjusted according to the actual solder paste. The data is only suitable for lead-free operation, refer to Figure 8-2 for lead-free reflow soldering operation guidelines.

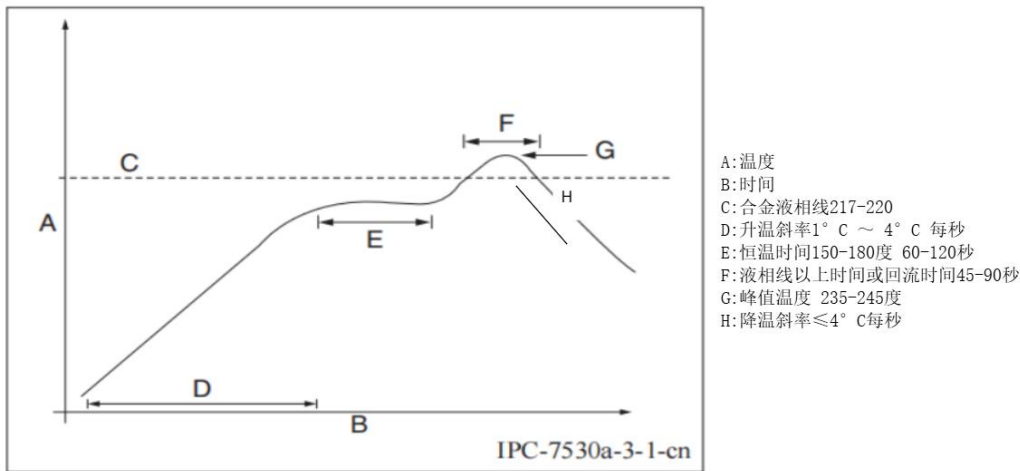


Figure 8.1 Lead-Free Reflow Soldering Operation Guide

Note

(1) During the production welding or any process that may directly contact the module, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) should be used to wipe the module shielding cover; otherwise, it may cause the shielding cover to rust.

(2) The Lierda module's white copper radium-carved shield can meet the following requirements: After 12 hours of neutral salt spray test, the radium-carved information is clear and identifiable, and the QR code can be scanned (there may be white rust).

(3) If you need to spray paint on the module, please make sure that the spray material used will not react chemically with the module shield or PCB, and also ensure that the spray material will not flow into the interior of the module.

(4) Please do not clean the module with ultrasonic waves, as it may cause damage to the internal crystals of the module.

(5) Due to the complexity of the SMT process, in case of uncertainties or processes not mentioned in the documentation (such as selective wave soldering, ultrasonic welding), please confirm with Lierda technical support before starting the SMT process.

8.1.3 Maintenance

If the module has defects such as virtual welding and short circuit that require repair, please follow the parameters below:

- ◆ Lead-free process: Soldering iron temperature $380 \pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$.
- ◆ Lead process: Soldering iron temperature $350 \pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$.

8.2 Packaging specifications

The factory packaging of this module is carried out using a rubber wheel method, the reference dimensions of the rubber wheel are as follows:

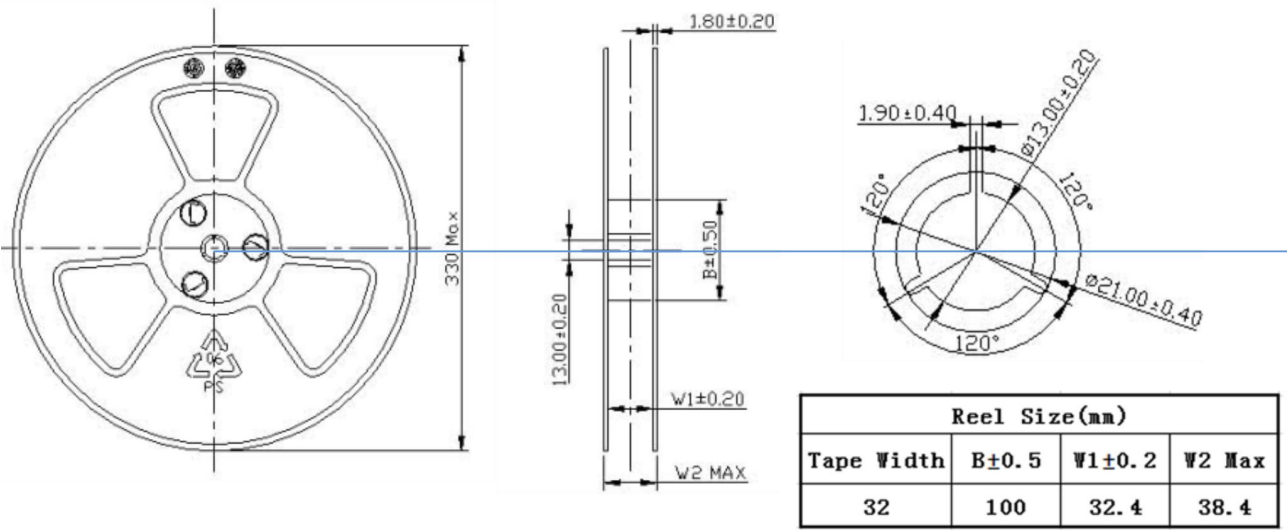


Figure 8.2 Dimensional drawing of the carrier roller

The loading direction is as follows: (Refer to the diagram, actual content may vary, pay attention to the position of module PIN1)

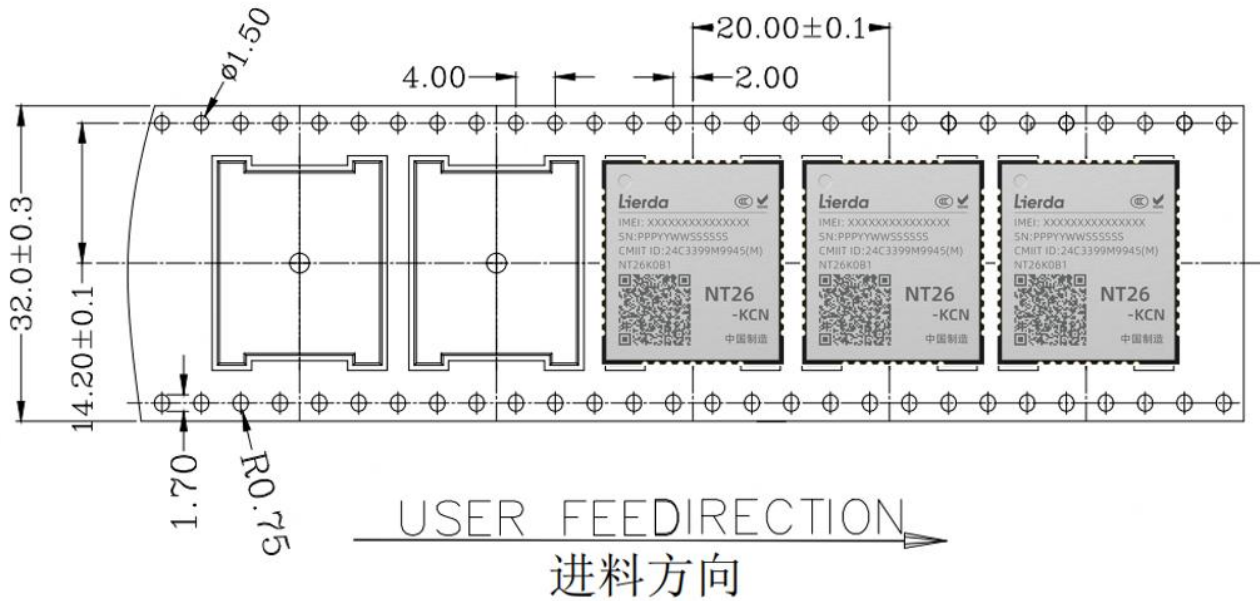


Figure 8.3 Loading direction indication diagram

8.3 Storage conditions

The module is shipped in the form of a vacuum-sealed bag. The humidity sensitivity level of the module is 3 (MSL 3), and its storage needs to follow the following conditions:

1. Recommended storage conditions: Temperature 23 ± 5 °C, and relative humidity 35~60%.
2. Under recommended storage conditions, the module can be stored in a vacuum-sealed bag for 12 months.

At a temperature of 23 ± 5 °C and relative humidity below 60% in the workshop, the module's workshop life after unpacking is 168 hours. Under these conditions, the module can be directly subjected to reflow production or other high-temperature operations. Otherwise, the module needs to be stored in an environment with relative humidity below 10% (e.g., a dry cabinet) to maintain its dryness.

If the module is in the following condition, it needs to be pre-baked to prevent the module from absorbing moisture and getting damp, which may cause PCB foaming, cracks, and delamination after high-temperature soldering:

- Storage temperature and humidity do not meet the recommended storage conditions;
- The module cannot be produced or stored according to the above clause 3 after unpacking;
- Vacuum packaging leaks, material scattered.
- Module before rework.

Module baking process:

- High temperature baking at 120 ± 5 °C for 8 hours is required;
- Modules that have undergone secondary baking must be welded within 24 hours after baking, otherwise they still need to be stored in the drying oven.

Notes

(1)To prevent and reduce the occurrence of poor welding such as blistering and delamination caused by moisture in the module, strict control should be carried out, and the vacuum package should not be opened for a long time exposure to air.

(2)Before baking, remove the original packaging, transfer the module to a high-temperature resistant carrier to prevent plastic trays or reels from deforming due to heat. If only short-term baking is needed, refer to the IPC/JEDEC J-STD-033 standard.

(3)When unpacking and transferring modules, anti-static measures must be taken, such as wearing anti-static gloves.

9 Related documents and terminology abbreviations

9.1 Related documents

Reference documents are as follows:

Table 9-1 Related Documents

Serial number	Document Name
[1]	Lierda NT26KxBx series Hardware Reference Design Manual
[2]	Lierda LTE Standard(L) Series AT Command Manual
[3]	Lierda LTE Standard (Q) Series AT Command Manual
[4]	IPC/JEDEC J-STD-033 specification

9.2 Term abbreviation

The abbreviations and their corresponding explanations mentioned in this document are listed in the table below:

Table 9-2 Term Abbreviations

Abbreviation	Full English name	Chinese full name
3GPP	3rd Generation Partnership Project	Third Generation Partner Program
ADC	Analog-to-Digital Converter	Analog to Digital Conversion
ANT	Antenna	Antenna
ASM	Antenna Switch Module	Antenna switch module
DAC	Digital -to- Analog Converter	Analog-to-Digital Conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter

DCXO	Digitally Controlled Crystal Oscillator	Numerically Controlled Crystal Oscillator
DRX	Discontinuous Reception	Non-continuous reception
DTE	Data Terminal Equipment	Data terminal equipment
ECL	Equivalent Class Level	Network coverage level
ESD	Electro-Static discharge	Electrostatic discharge
EOS	Electrical Overtress	Electrical super stress (surge)
ESR	Equivalent Series Resistance	Equivalent series resistance
EVK	Evaluation Kit	Assessment Toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	General Input Output
I/O	Input/Output	Input and output interface
I _{max}	Maximum Load Current	Maximum current
I _{norm}	Normal Current	Normal (typical) current
bps	Bits Per Second	Units of speed
LCC	Leadless Chip Carriers	N/A lead frame chip carrier package
LCM	LCD Module	LCD display module
LDO	Low Dropout Regulator	Low dropout linear regulator
LGA	Land Grid Array	Grid array packaging
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M protocol
MCU	Micro Controller Unit	Microcontrol unit
MSL	Moisture Sensitivity levels	Wet sensitivity level
PCB	Printed Circuit Board	Printed Circuit Board
PCBA	Printed Circuit Board Assembly	Printed circuit board

		components
PMU	Power Management Unit	Power Management Unit
PSM	Power Saving Mode	Energy saving mode
RF	Radio Frequency	Radio Frequency
RoHS	Restriction of Hazardous Substances	Limitation of harmful substances
RX	Receive	Receive
TAU	Tracking Area Update	Tracking area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient Voltage Suppression Diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	Universal asynchronous receiver-transmitter and transmitter
UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
URC	Unsolicited Result Code	Non-request result code
(U)SIM	(Universal) Subscriber Identification Module	User Identity Recognition Module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (typical) voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level

Vlmax	Absolute Maximum Input Voltage Value	Maximum input level
Vlmin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level
VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level